Advanced Uncertainty Quantification Notes

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1. Representation of Second Order Random Fields

1.1. Essential Revision

Definition 1.1 (Probability space)**.** A probability space contains 3 components

$$(\Omega, \mathcal{F}, \mathbf{P}).$$

- Ω: a sample space (a set)
- $\mathcal{F}$: a σ-algebra (a collection of subsets of Ω)
- $\mathbf{P}$: a (probability) measure

Definition 1.2 (Real-valued random variable)**.** A random variable X is a function which defines on the probability space.

$$X : \Omega \to \mathbf{R}.$$

***Example* 1.3.** Some real-valued random variable examples:

$$X \sim N(0,1) \text{ (Normal)}, \quad X \sim U(-1,1) \text{ (Uniform)}.$$

Definition 1.4 ($\mathbf{R}^N$-valued random varible)**.** A $\mathbf{R}^N$-valued random varible $\boldsymbol{X}$ is a function

$$\boldsymbol{X} : \Omega \to \mathbf{R}^N \text{ where } \boldsymbol{X} = (X_1, \ldots, X_N)^\top.$$

***Example* 1.5.**

$$\boldsymbol{X} \sim N(\boldsymbol{\mu}, C) \text{ (multivariate Normal)}$$

Definition 1.6 (Stochastic Process)**.** Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a **probability space** and let $\mathbf{T} \subset \mathbf{R}$. A real-valued *stochastic process* $f(t, \omega)$ can be viewed as

- A real-valued **function** $f : \mathbf{T} \times \Omega \to \mathbf{R}$.
- A collection of real-valued **random variables** for each $t \in \mathbf{T}$

$$\{f(t, \cdot), t \in \mathbf{T}\}.$$

- A collection of **realisations**

$$\{f(\cdot, \omega), \omega \in \Omega\}.$$

($\star$) We will write $f(t, \omega)$ to stress that a stochastic process is a *function*.

1.2. Motivating Example

***Example* 1.7** (Deterministic ODE)**.** Let $D = (0,1)$ and consider the following boundary value problem (BVP):

Find $u : \overline{D} \to \mathbf{R}$ such that

$$-\frac{\mathrm{d}}{\mathrm{d}x}\left(a(x)\frac{\mathrm{d}u(x)}{\mathrm{d}x}\right) = 1, \quad x \in D,$$

together with

$$u(0) = \alpha, \quad u(1) = \beta.$$

This is a simple model of heat diffusion in a wire where

- x denotes **spatial position**.

- $u(x)$ represents the temperature.
- $a(x)$ is the thermal conductivity (or diffusion coefficient).

If $a = 1$, and $\alpha = \beta = 0$, then $u(x) = x(1-x)/2$ (analytic solution is available).

1.2.1. Random diffusion coefficient

Suppose a is not a simple function of x, but a *stochastic process* (also known as a *random field*).

Let $(\Omega, \mathcal{F}, \mathbf{R})$ be a probability space, and let $\boldsymbol{\xi}_1, \xi_2, \ldots, \xi_{50} : \Omega \to \mathbf{R}$ be **independent**. Now define

$$a(x,\omega) = 1 + \sum_{i=1}^{50} \frac{4}{i^2\pi^2} \cos(i\pi x)\xi_i(\omega), \quad x \in D, \quad \omega \in \Omega. \tag{1.1}$$

In particular, suppose $\xi_1, \xi_2, \ldots, \xi_{50} \sim U(-1,1)$. We could then generate 20 realisations of $a(\cdot, \omega_j)$ shown in figure 1 .

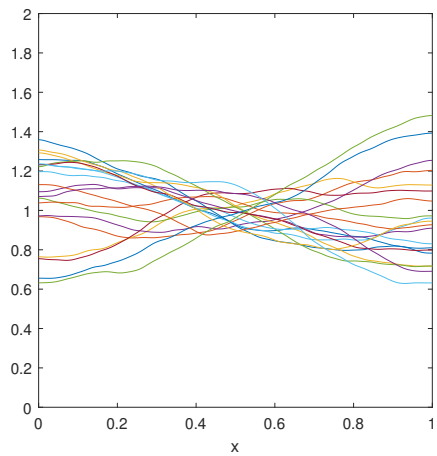

FIGURE 1: **Twenty realisations** $a(\cdot, \omega_j)$ of the diffusion coefficient for $j = 1, 2, \ldots, 20$.

1.2.2. Stochastic ODE Model

Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and let $D = (0,1)$ (**the spatial domain**).

Find $u : \overline{D} \times \Omega \to \mathbf{R}$ such that **P**-a.s,

$$-\frac{\mathrm{d}}{\mathrm{d}x}\left(a(x,\omega)\frac{\mathrm{d}u(x,\omega)}{\mathrm{d}x}\right) = 1, \quad x \in D,$$

and

$$u(0,\omega) = \alpha, \quad u(1,\omega) = \beta.$$

Question.

- Does a solution $u(x,\omega)$ exist? Is it unique?
- What kind of random input $a(x,\omega)$ is allowed?
- If a solution $u(x,\omega)$ exist, what are its properties?
- What *function space* does $u(x,\omega)$ belong to?

1.2.3. Forward UQ

Given an input random field $a(x,\omega)$, estimate statistical quantity of interest associated with $u(x,\omega)$.

We can use *Monte Carlo* with FD approximation, to estimate the **mean**

$$\mathbf{E}[u(x,\omega)] \approx \frac{1}{M}\sum_{j=1}^{M} u(x,\omega_j) \approx \frac{1}{M}\sum_{j=1}^{M} u_h^j(x) =: \mu_{M,h}.$$

And similarly the variance

$$\mathbf{Var}[u(x,\omega)] \approx \frac{1}{M-1}\sum_{j=1}^{M}(u_h^j(x)-\mu_{M,h})^2.$$

Question. How can we **generate realisations** of the random input $a(x,\omega)$. (Discussed in section 2)

1.3. Random Variables

In UQ1, we met $\mathbf{R}$-valued and $\mathbf{R}^N$-valued random variables,

$$X : \Omega \to \mathbf{R}, \quad \boldsymbol{X} : \Omega \to \mathbf{R}^N,$$

we will make this concept **more general**.

Definition 1.8 (Measure Space)**.** Let Ψ be a set with a σ-algebra $\mathcal{G}$. We call $(\Psi,\mathcal{G})$ a *measurable space* and each $G \in \mathcal{G}$ is a *measurable set*.

Given a *measure* π on $(\Psi,\mathcal{G})$, we call $(\Psi,\mathcal{G},\pi)$ a *measure space*.

***Example* 1.9.** A probability space $(\Omega,\mathcal{F},\mathbf{P})$ is a measure space, with

$$\mathbf{P}(\Omega) = 1.$$

Definition 1.10 (Borel sigma-algebra)**.** The *Borel sigma-algebra* $\mathcal{B}(D)$ is defined to be the **smallest** sigma-algebra containing all **open subsets** of D.

***Example* 1.11.** Let $\Psi = D$, where $D \subset \mathbf{R}$. $(D,\mathcal{B}(D))$ is a measurable space.

For each $(a,b) \in \mathcal{B}(D)$, we can assign a 'size', using *Lebesgue measure*,

$$Leb(a,b) := |b-a|.$$

and $(D,\mathcal{B}(D),Leb)$ is a *measure space*.

The measure of a Borel-set can be represented as an **integral**

$$Leb(a,b) = \int_a^b 1\mathrm{d}Leb(x) = \int_a^b 1\mathrm{d}x.$$

Definition 1.12 (Random Variable)**.** Let $(\Omega,\mathcal{F},\mathbf{P})$ be a probability space and let $(\Psi,\mathcal{G})$ be a measurable space. $X : \Omega \to \Psi$ is a Ψ-valued random variable, if it is $\mathcal{F}$*-measurable*. That is,

$$\forall G \in \mathcal{G} \quad X^{-1}(G) = \{\omega \in \Omega | X(\omega) \in G\} \in \mathcal{F}. \tag{1.2}$$

Definition 1.13 (Probability distribution)**.** The **probability distribution** of X is the *measure* $\mathbf{P}_X$ on $(\Psi,\mathcal{G})$ defined by

$$\mathbf{P}_X(G) := \mathbf{P}(X^{-1}(G)), \quad \forall G \in \mathcal{G},$$

and $(\Psi,\mathcal{G},\mathbf{P}_X)$ is also a **probability space**.

1.3.1. Continuous Random Variables

We will only be dealing with random variables where Ψ is an *infinite set* like $\mathbf{R}$, or a set of functions on a spatial domain D.

Measurability is an important concept that allows us to define integration.

For continuous random variables $X : \Omega \to \Psi$,

$$\mathbf{P}(\{\omega \in \Omega | X(\omega) \in G\}) = \mathbf{P}_X(G) = \int_G 1 \mathrm{d}\mathbf{P}_X(x). \tag{1.3}$$

And the expectation:

$$\mathbf{E}[X] := \int_\Omega X(\omega) \mathrm{d}\mathbf{P}(\omega) = \int_\Psi x \ \mathrm{d}\mathbf{P}_X(x). \tag{1.4}$$

1.3.2. Real-valued Random Variable ($\Psi = \mathbf{R}$)

If we know the *probability density function* $\rho(x)$ associated with $\mathbf{P}_X$, we can re-write the integrals w.r.t. $\mathbf{P}_X$ as standard integrals w.r.t. **Lebesgue measure**.

Probability from equation (3.35):

$$\mathbf{P}(\{\omega \in \Omega | a < X(\omega) < b\}) = \mathbf{P}_X(a, b) = \int_a^b 1 \mathrm{d}\mathbf{P}_X(x) = \int_a^b \rho(x) \mathrm{d}x. \tag{1.5}$$

Expectation from equation (3.36)

$$\mathbf{E}[X] := \int_\Omega X(\omega) \mathrm{d}\mathbf{P}(\omega) = \int_\mathbf{R} x \ \mathrm{d}\mathbf{P}_X(x) = \int_\mathbf{R} x\rho(x) \mathrm{d}x. \tag{1.6}$$

1.4. Random Fields

Definition 1.14 (Random Field)**.** Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and let $D \subset \mathbf{R}^d$. A real-valued *random field* $u : D \times \Omega \to \mathbf{R}$ is a real-valued random variable for each $\boldsymbol{x} \in D$.

If $d = 1$ and $D \in \mathbf{R}$, then we call a random field (RF) a *stochastic process*.

Three interpretations:

- A real-valued **function** $u : D \times \Omega \to \mathbf{R}$.
- A collection of real-valued **random variables** for each $\boldsymbol{x} \in D$

$$\{u(\boldsymbol{x}, \cdot), \ \boldsymbol{x} \in D\}.$$

- A collection of **realisations** $\{u(\cdot, \omega), \omega \in \Omega\}$.

***Example* 1.15** ($d = 1$)**.** Let $D = [0, 2\pi]$ and consider

$$a(x, \omega) = \cos(x)\xi_1(\omega) + \sin(x)\xi_2(\omega), \quad x \in D, \quad \xi_1, \xi_2 \sim N(0, 1) \ iid.$$

Question. What kind of functions are the realisations?

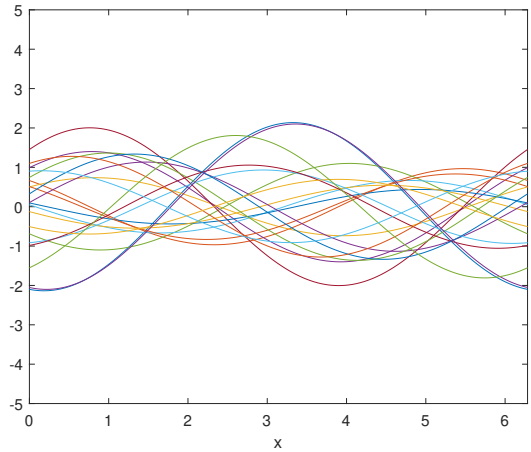

FIGURE 2: Twenty *realisations* $a(x, \omega_j)$, $j = 1, 2, \ldots, 20$ are plotted above.

1.4.1. Square-integrable functions

Realisations of the RF in this example are **continuous functions** of $x \in D$. They are also **differentiable**. *This is not true in general! (Brownian motion).*

We will work with RFs with realisations which are at least *square-integrable.*

Definition 1.16 ($L^2(D)$ space)**.** Let $D \subset \mathbf{R}^d$. $L^2(D)$ is the set of Borel-measurable functions $u : D \to \mathbf{R}$ such that

$$\|u\|_{L^2(D)} := \left(\int_D |u(\boldsymbol{x})|^2 \mathrm{d}\boldsymbol{x} \right)^{1/2} < \infty. \tag{1.7}$$

This is a *Hilbert Space*, which is equipped with an *inner-product*

$$\langle u, v \rangle := \int_D u(\boldsymbol{x}) v(\boldsymbol{x}) \mathrm{d}\boldsymbol{x}, \tag{1.8}$$

where $\|u\|_{L^2(D)} = \sqrt{\langle u, u \rangle}$.

***Example* 1.17** (Example 1.15 revisit)**.** Let $D = [0, 2\pi]$ and consider

$$a(x, \omega) = \cos(x)\xi_1(\omega) + \sin(x)\xi_2(\omega), \quad x \in D, \quad \xi_1, \xi_2 \sim N(0, 1) \text{ } iid.$$

For each $x \in D$, $a(x, \cdot)$ is a **random variable**. For any $x \in D$,

$$\mathbf{E}[a(x, \cdot)] = \cos(x)\,\mathbf{E}[\xi_1] + \sin(x)\,\mathbf{E}[\xi_2] = 0 + 0 = 0, \tag{1.9}$$

so it $a(x, \omega)$ clearly **mean zero**.

1.4.2. Finite Second Moment

Definition 1.18 ($L^2(\Omega)$ space)**.** Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space. $L^2(\Omega)$ is the set of $\mathcal{F}$-measurable functions $X : \Omega \to \mathbf{R}$ (i.e., real-valued random variables) such that

$$\|X\|_{L^2(\Omega)} := \mathbf{E}[X^2]^{1/2} < \infty. \tag{1.10}$$

This is also a *Hilbert Space*, which is equipped with the *inner-product*

$$\langle X, Y \rangle := \mathbf{E}[XY]. \tag{1.11}$$

Recall that

$$\mathbf{E}[X^2] := \int_\Omega X(\omega)^2 \mathrm{d}\mathbf{P}(\omega) = \int_{\mathbf{R}} x^2 \mathrm{d}\mathbf{P}_X(x) = \int_{\mathbf{R}} x^2 \rho(x) \mathrm{d}x. \tag{1.12}$$

Random variables in $L^2(\Omega)$ have *finite second moment.*

1.5. Second-Order Random Fields

Definition 1.19 (Second-Order RF)**.** Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and let $D \subset \mathbf{R}^d$. A real-valued RF $a(\boldsymbol{x}, \omega)$ is *second order* if $a(\boldsymbol{x}, \cdot) \in L^2(\Omega) \quad \forall \boldsymbol{x} \in D$.

Such RFs have well-defined **mean** and **covariance** functions.

The *mean* function $\mu : D \to \mathbf{R}$ is defined by

$$\mu(\boldsymbol{x}) := \mathbf{E}[a(\boldsymbol{x}, \omega)] = \int_\Omega a(\boldsymbol{x}, \omega) \mathrm{d}\mathbf{P}(\omega). \tag{1.13}$$

The *covariance* function $C : D \times D \to \mathbf{R}$ is defined by

$$C(\boldsymbol{x}, \boldsymbol{y}) := \mathrm{Cov}(a(\boldsymbol{x}, \omega), a(\boldsymbol{y}, \omega)), \quad \boldsymbol{x}, \boldsymbol{y} \in D. \tag{1.14}$$

Using the definition of covariance

$$C(\boldsymbol{x}, \boldsymbol{y}) = \mathbf{E}[(a(\boldsymbol{x}, \omega) - \mu(\boldsymbol{x}))(a(\boldsymbol{y}, \omega) - \mu(\boldsymbol{y}))]. \tag{1.15}$$

Given the covariance function $C(\boldsymbol{x}, \boldsymbol{y})$, the *variance* of a RF is

$$\mathbf{Var}[a(\boldsymbol{x}, \omega)] := C(\boldsymbol{x}, \boldsymbol{x}) = \mathbf{E}[(a(\boldsymbol{x}, \omega) - \mu(\boldsymbol{x}))^2]. \tag{1.16}$$

1.5.1. Valid Covariance functions

I mportant: Not all functions $C(\boldsymbol{x}, \boldsymbol{y})$ are valid covariance functions.

Theorem 1.20 (Covariance Functions)**.** *Let $D \in \mathbf{R}^d$ and let $a(\boldsymbol{x}, \omega)$ be a second-order RF. The covariance function $C : D \times D \to \mathbf{R}$ is* symmetric,*i.e.,*

$$C(\boldsymbol{x}, \boldsymbol{y}) = C(\boldsymbol{y}, \boldsymbol{x}) \quad \boldsymbol{x}, \boldsymbol{y} \in D$$

and non-negative definite*. That is, for any $N \in \mathbf{N}$ and any constants $\alpha_1, \ldots, \alpha_N \in \mathbf{R}$, and any points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N \in D$,*

$$\sum_{i=1}^N \sum_{j=1}^N \alpha_i C(\boldsymbol{x}_i, \boldsymbol{x}_j) \alpha_j \geq 0.$$

***Example* 1.21** (Covariance Functions)**.** Often, $C(\boldsymbol{x}, \boldsymbol{y})$ is a function of the *distance* between a pair of points $\boldsymbol{x}, \boldsymbol{y} \in D$.

Some examples for one-dimensional domains $D \subset \mathbf{R}$:

- The *exponential* covariance function is

$$C(x, y) = \sigma^2 \exp\left(\frac{-|x - y|}{\ell}\right)$$

  Here, $C(x, x) = \sigma^2$ is the (constant) variance and ℓ is the **correlation length**.

- The *Gaussian* covariance function ($d = 1$) is

$$C(x, y) = \sigma^2 \exp\left(\frac{-|x - y|^2}{\ell}\right).$$

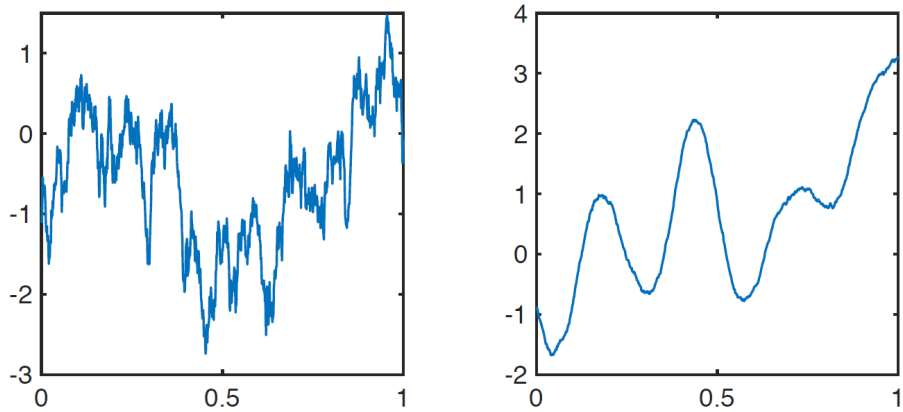

FIGURE 3: A **realisation** of mean zero RF with (Left) the *exponential* covariance function and (Right) the *Gaussian* covariance function with $\sigma = 1,\ \ell = 0.1$.

1.5.2. Smoothness

The smoothness of the **realisations** of a RF depends on the smoothness of the **covariance function**.

1.6. Gaussian Random Fields

To specify a RF, it is not enough to specify the domain D, the mean function $\mu(\boldsymbol{x})$ and covariance function $C(\boldsymbol{x}, \boldsymbol{y})$. We also have to specify the *probability distribution*.

Definition 1.22 (Gaussian RF)**.** Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space and let $D \subset \mathbf{R}^d$. A real-vallued RF $a(\boldsymbol{x}, \omega)$ with mean $\mu(\boldsymbol{x})$ and covariance $C(\boldsymbol{x}, \boldsymbol{y})$ is *Gaussian* if $\forall \boldsymbol{x}_1, \ldots, \boldsymbol{x}_N \in D$ (for any $N \in \mathbf{N}$),

$$\boldsymbol{a}(\omega) = [a(\boldsymbol{x}_1, \omega), \ldots, a(\boldsymbol{x}_N, \omega)]^\top \sim N(\boldsymbol{\mu}, C_N)$$

(i.e., **multivariate Normal**) with *mean vector*

$$\boldsymbol{\mu} \in \mathbf{R}^N, \quad [\boldsymbol{\mu}]_i = \mu(\boldsymbol{x}_i), \quad i = 1, \ldots, N,$$

and *covariance matrix*

$$C_N \in \mathbf{R}^{N \times N}, \quad [C_N]_{i,j} = C(\boldsymbol{x}_i, \boldsymbol{x}_j), \quad i, j = 1, \ldots, N.$$

***Example* 1.23** (Brownian motion)**.** Some Gaussian RFs have special name.

Let $\mathbf{T} \subset \mathbf{R}^+$. *Brownian Motion* $W : \mathbf{T} \times \Omega \to \mathbf{R}$ is a Gaussian stochastic process $W(t, \omega)$ with continuous realisations with

$$\mu(t) = 0, \quad C(s, t) = \min\{s, t\}.$$

***Example* 1.24** (Brownian Bridge)**.** The *Brownian Bridge* is a version of Brownian motion on a bounded domain that also satisfies **boundary conditions**.

Let $\mathbf{T} = [0, 1]$. The **Brownian Bridge** $B : \mathbf{T} \times \Omega \to \mathbf{R}$ is a Gaussian stochastic process $B(t, \omega)$ satisfying

$$B(0, \omega) = 0, \quad B(1, \omega) = 0$$

almost surely, with

$$\mu(t) = 0, \quad C(s, t) = \min\{s, t\} - st.$$

1.7. Intergal Operators and Covariance Functions

1.7.1. Square-integrable functions on $D \times D$

We previously defined the space $L^2(D)$ of *square-integrable* functions $u : D \to \mathbf{R}$.

Definition 1.25 ($L^2(D \times D)$ space)**.** Let $D \in \mathbf{R}^d$. A function $G : D \times D \to \mathbf{R}$ belongs to $L^2(D \times D)$ if

$$\|G\|_{L^2(D\times D)} := \left(\int_D \int_D G(\boldsymbol{x}, \boldsymbol{y})^2 \mathrm{d}\boldsymbol{x}\mathrm{d}\boldsymbol{y}\right)^{1/2} < \infty. \tag{1.17}$$

Definition 1.26 (Integral Operator with Kernel G)**.** Let $D \subset \mathbf{R}^d$. Given a 'kernel' $G \in L^2(D \times D)$, the *integral operator* $\mathcal{L} : L^2(D) \to L^2(D)$ associated with G is defined by

$$\mathcal{L}f(\boldsymbol{x}) := \int_D G(\boldsymbol{x}, \boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{y}, \quad f \in L^2(D). \tag{1.18}$$

Such operators are: *bounded, linear* and *compact* on $L^2(D)$.

Let $H = L^2(D)$.

Definition 1.27 (Linear operator)**.** We say $\mathcal{L} : H \to H$ is *linear* on H if

- $\mathcal{L}(f + g) = \mathcal{L}(f) + \mathcal{L}(g), \quad \forall f, g \in H$
- $\mathcal{L}(\alpha f) = \alpha \mathcal{L}(f), \quad \forall \alpha \in \mathbf{R}$ and $\forall f \in H$.

Definition 1.28 (Bounded operator)**.** We say $\mathcal{L} : H \to H$ is *bounded* on H if there exist a constant $K > 0$ such that

$$\|\mathcal{L}f\|_H \le K\|f\|_H, \quad \forall f \in H.$$

Theorem 1.29 (Symmetric operator)**.** *If the kernel G is s ymmetric*

$$G(\boldsymbol{x}, \boldsymbol{y}) = G(\boldsymbol{y}, \boldsymbol{x}), \quad \boldsymbol{x}, \boldsymbol{y} \in D,$$

then the operator $\mathcal{L}$ in definition 1.26 is also s ymmetric, i.e.,

$$\langle \mathcal{L}f, g\rangle = \langle f, \mathcal{L}g\rangle, \quad \forall f, g \in L^2(D),$$

where $\langle \cdot, \cdot \rangle$ is the inner-product on $L^2(D)$.

Recall. Covariance functions $C(\boldsymbol{x}, \boldsymbol{y})$ are always s ymmetric. Given a covariance function $C(\boldsymbol{x}, \boldsymbol{y})$ that belongs to $L^2(D \times D)$, we can use definition 1.26 to construct an *integral operator*

$$\mathcal{C} : L^2(D) \to L^2(D)$$

through the relation

$$\mathcal{C}f(\boldsymbol{x}) := \int_D C(\boldsymbol{x}, \boldsymbol{y}) f(\boldsymbol{y}) \mathrm{d}\boldsymbol{y}, \quad f \in L^2(D),$$

this is *linear, bounded, compact* and *symmetric*.

1.7.2. Eigenvalues & Eigenfunctions

There are infinitely many *eigenvalues* λ and *eigenfunctions* $\phi(\neq 0)$ satisfying

$$\mathcal{C}\phi(\boldsymbol{x}) = \lambda\phi(\boldsymbol{x}).$$

Using the definition of $\mathcal{C}$, we have

$$\int_D C(\boldsymbol{x}, \boldsymbol{y})\phi(\boldsymbol{y})\mathrm{d}\boldsymbol{y} = \lambda\phi(\boldsymbol{x}).$$

***Example* 1.30** (Brownian Bridge)**.** Let $\mathbf{T} = [0,1]$. Recall that the *Brownian Bridge* $B : \mathbf{T} \times \Omega \to \mathbf{R}$ is Gaussian stochastic process with covariance function

$$C(s,t) = \min\{s,t\} - st.$$

The associated integral operator is defined via

$$\mathcal{C}f(t) := \int_0^1 C(s,t)f(s)\mathrm{d}s = \int_0^1 (\min\{s,t\} - st)f(s)\mathrm{d}s, \quad f \in L^2(\mathbf{T}).$$

The associated eigenvalues and eigenfunctions satisfies $\mathcal{C}\phi(t) = \lambda\phi(t)$, or

$$\int_0^1 (\min\{s,t\} - st)\phi(s) = \lambda\phi(t). \tag{1.19}$$

Notice that

$$\begin{aligned}\mathcal{C}f(t) = \int_0^1 (\min\{s,t\} - st)f(s)\mathrm{d}s &= \int_0^t (s - st)f(s)\mathrm{d}s + \int_t^1 (t - st)f(s)\mathrm{d}s \\ &= (1-t)\int_0^t sf(s)\mathrm{d}s + t\int_t^1 (1-s)f(s)\mathrm{d}s.\end{aligned} \tag{1.20}$$

Thus the eigenfunction ϕ and eigenvalues λ can be determined by solving

$$(1-t)\int_0^t s\phi(s)\mathrm{d}s + t\int_t^1 (1-s)\phi(s)\mathrm{d}s = \lambda\phi(t)$$

Notice that $\phi(0) = 0, \phi(1) = 0$. Differentiate w.r.t. t, we have

$$\begin{aligned}t\phi(t) - t^2\phi(t) - \int_0^t s\phi(s)\mathrm{d}s + \int_t^1 (1-s)\phi(s)\mathrm{d}s - t(1-t)\phi(t) &= \lambda\phi'(t) \\ -\int_0^t s\phi(s)\mathrm{d}s + \int_t^1 (1-s)\phi(s)\mathrm{d}s &= \lambda\phi'(t) \\ -t\phi(t) + (-(1-t)\phi(t)) &= \lambda\phi''(t) \\ \lambda\phi''(t) + \phi(t) &= 0 \\ \phi(t) + \frac{1}{\lambda}\phi(t) = 0 \Leftrightarrow \quad &\underbrace{\phi''(t) + \nu\phi(t) = 0}_{\text{Second Order ODE with } \nu = 1/\lambda}\end{aligned} \tag{1.21}$$

We then have the general solution:

$$\phi(t) = A\cos(\omega t) + B\sin(\omega t), \quad \phi(0) = \phi(1) = 0.$$

This system will gives

$$\phi_i(t) = B_i \sin(i\pi t), \quad \lambda_i = \frac{1}{\omega^2} = \frac{1}{i^2\pi^2}, \quad i \in \mathbf{Z}^+.$$

We could normalize this function by acquiring $\|\phi_i\|_{L^2(0,1)} = 1$, and we have $B_i = \sqrt{2}$.

Theorem 1.31 (Hilbert-Schmidt Spectral Theorem)**.** *Let $H = L^2(D)$, and $\mathcal{L} : H \to H$ be a bounded linear operator that is symmetric and compact. Denote the eigenvalue of $\mathcal{L}$ by λ_i, ordered such that $|\lambda_i| \le |\lambda_{i+1}|$, and denoted the corresponding eigenfunctions be $\phi_i \in L^2(D)$. Then*

- *The eigenvalues λ_i are real and $\lambda_i \to 0$ as $i \to \infty$.*
- *The eigenfunctions ϕ_j can be chosen to form an* orthonormal *basis for the range of $\mathcal{L}(L^2(D))$.*

- *For any* $u \in L^2(D)$,
$$\mathcal{L}u = \sum_{i=1}^{\infty} \lambda_i \langle u, \phi_i \rangle_{L^2(D)} \phi_i.$$

1.8. Karhuren-Loève expansion

Definition 1.32 ($L^2(\Omega, L^2(D))$ space)**.** Let $(\Omega, \mathcal{F}, \mathbf{P})$ be a probability space, and let $D \in \mathbf{R}^d$. $L^2(\Omega, L^2(D))$ is the space of $L^2(D)$-valued random variables $a : \Omega \to L^2(D)$ satisfying

$$\|a\|_{L^2(\Omega, L^2(D))} := \mathbf{E}[\|a(\boldsymbol{x}, \omega)\|^2_{L^2(D)}]^{1/2} \leq \infty.$$

We can interpret such functions as RF: If $a(\boldsymbol{x}, \omega) \in L^2(\Omega, L^2(D))$.

- For each ω, we have a realisation $a(\cdot, \omega) \in L^2(D)$.
- For each $\boldsymbol{x}$, we have a random variable $a(\boldsymbol{x}, \cdot) \in L^2(\Omega)$.

1.8.1. Series Expansion

Suppose $a(\boldsymbol{x}, \omega) \in L^2(\Omega, L^2(D))$. If we know a *basis* $\{\phi_i(\boldsymbol{x}), i = 1, 2, \ldots, \}$ for $L^2(D)$, then any *realisation* of $a(\boldsymbol{x}, \omega)$ can be represented as a linear combination of those functions $a(\boldsymbol{x}, \omega) = \sum_{i=1}^{\infty} a_i(\omega)\phi_i(\boldsymbol{x})$, where the coefficients are random variables. The KL expansion uses the eigenfunctions ϕ_i of the integral operator associated with the covariance function as the chosen basis.

Theorem 1.33 (Karhuren-Loève expansion)**.** *Suppose* $a(\boldsymbol{x}, \omega) \in L^2(\Omega, L^2(D))$. *Then*

$$a(\boldsymbol{x}, \omega) = \underbrace{\mu(\boldsymbol{x})}_{\textit{Not random}} + \sum_{i=1}^{\infty} \sqrt{\lambda_i} \underbrace{\xi_i(\omega)}_{\textit{Random variables}} \phi_i(\boldsymbol{x}). \tag{1.22}$$

- $\mu(\boldsymbol{x})$ *is the mean function.*
- $\{\lambda_i, \phi_i(\boldsymbol{x})\}$ *are the eigenvalues and normalized eigenfunctions of the integal operator* $\mathcal{C}$ *associated with the covariance function where* $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$.
- *The random variables are*
$$\xi(\omega) := \frac{1}{\sqrt{\lambda_i}} \langle a(\boldsymbol{x}, \omega) - \mu(\boldsymbol{x}), \phi_i(\boldsymbol{x}) \rangle,$$
and they have mean zero, variance one, and are uncorrelated.
- *The series converges in the* $L^2(\Omega, L^2(D))$ *sense.*

Remark 1.34. The distribution of the random variables ξ_i appearing in KL expansion depends on the distribution of the RF $a(\boldsymbol{x}, \omega)$.

- If $a(\boldsymbol{x}, \omega)$ is a *Gaussian RF*, then the ξ_i are also Gaussian.
- Uncorrelated Gaussian random variables are independent.

1.8.2. Convergence

Note that the function in $L^2(D)$ does not need to be continuous. When we write $a(\boldsymbol{x}, \omega) = \mu(\boldsymbol{x}) + \sum_{i=1}^{\infty} \sqrt{\lambda_i}\xi_i(\omega)\phi_i(\boldsymbol{x})$, this does not mean that the series converges to $a(\boldsymbol{x}, \omega)$ for all $\boldsymbol{x} \in D$. But we could

have converges in 'the $L^2(\Omega, L^2(D))$ sense', which means that if we define

$$a_J(\boldsymbol{x}, \omega) := \mu(\boldsymbol{x}) + \sum_{i=1}^{J} \sqrt{\lambda_i}\xi_i(\omega)\phi_i(\boldsymbol{x}),$$

then

$$\|a(\boldsymbol{x}, \omega) - a_J(\boldsymbol{x}, \omega)\|_{L^2(\Omega, L^2(D))} \to 0 \quad \text{as} \quad J \to \infty.$$

1.9. Truncated KL expansion

What we defined previously

$$a_J(\boldsymbol{x}, \omega) := \mu(\boldsymbol{x}) + \sum_{i=1}^{J} \sqrt{\lambda_i}\xi_i(\omega)\phi_i(\boldsymbol{x}),$$

is the *truncated expansion* which is often used as an approximation to do numerical approximations.

***Example* 1.35** (Browinan Bridge)**.** Let $\mathbf{T} = [0, 1]$, recall that the Browian Bridge $B : \mathbf{T} \times \Omega \to \mathbf{R}$ is a Gaussian RF (stochastic process) with $\mu(t) = 0$, $C(s, t) = \min\{s, t\} - st$. The eigenvalues and *normalised* eigenfunctions of the integral operator are

$$\lambda_i = \frac{1}{i^2\pi^2}, \quad \phi_i(t) = \sqrt{2}\sin(i\pi t), \quad i = 1, 2, \ldots.$$

Using the definition of KL expansion:

$$B(t, \omega) = \sum_{i=1}^{\infty} \frac{\sqrt{2}}{i\pi}\xi_i(\omega)\sin(i\pi t), \quad \xi_i \sim N(0, 1) iid.$$

we could define the truncated KL expansion for $B(t, \omega)$,

$$B_J(t, \omega) = \sum_{i=1}^{J} \frac{\sqrt{2}}{i\pi}\xi_i(\omega)\sin(i\pi t), \quad \xi_i \sim N(0, 1) iid.$$

From figure 4, we could see the series is converging.

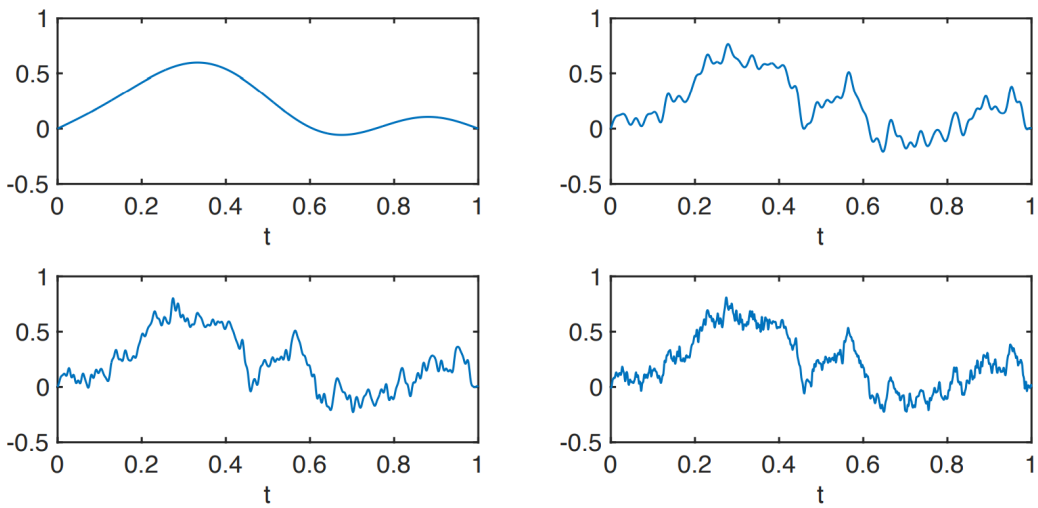

FIGURE 4: One realisation $B_J(t, \omega^*)$ of the truncated KL expansion of the standard Brownian bridge $B(t, \omega)$ defined in Example 1.30 for $J = 5, 50, 100, 500$ (left to right, top to bottom).

1.9.1. Error in Truncated KL expansion

Theorem 1.36. *If $a(\boldsymbol{x},\omega) \in L^2(\Omega, L^2(D))$, and $a_J(\boldsymbol{x},\omega)$ is the truncated approximation to $a(\boldsymbol{x},\omega)$. Then*

$$\mathbf{E}[\|a(\boldsymbol{x},\omega) - a_J(\boldsymbol{x},\omega)\|^2_{L^2(D)}] = \sum_{i=J+1}^{\infty} \lambda_i$$

Theorem 1.37 (Mercer's Theorem)**.** *Let $a(\boldsymbol{x},\omega) \in L^2(\Omega, L^2(D))$ be a second-order RF with KL expansion:*

$$a(\boldsymbol{x},\omega) = \mu(\boldsymbol{x}) + \sum_{i=1}^{\infty} \sqrt{\lambda_i}\xi_i(\omega)\phi_i(\boldsymbol{x}).$$

If $D \subset \mathbf{R}^d$ is bounded and closed and if $C(\boldsymbol{x},\boldsymbol{y})$ is continuous, then

- *The eigenfunctions $\phi_i(\boldsymbol{x})$ are continuous.*
- *The covariance function can be expressed as*

$$C(\boldsymbol{x},\boldsymbol{y}) = \sum_{i=1}^{\infty} \lambda_i \phi_i(\boldsymbol{x})\phi_j(\boldsymbol{y})$$

where the series converges uniformly*. That is :*

$$\|C - \widehat{C}\|_\infty = \sup_{\boldsymbol{x},\boldsymbol{y}\in D} |C(\boldsymbol{x},\boldsymbol{y}) - \widehat{C}(\boldsymbol{x},\boldsymbol{y})| \to 0, \quad \text{as} \quad J \to \infty,$$

where $\widehat{C}(\boldsymbol{x},\boldsymbol{y})$ is the truncated convariance function approximation.

1.9.2. Error in Variance

It can be shown that $\widehat{C}(\boldsymbol{x},\boldsymbol{y})$ is covariance function of the truncated RF

$$a_J(\boldsymbol{x},\omega) = \mu(\boldsymbol{x}) + \sum_{i=1}^{J} \sqrt{\lambda_i}\xi_i(\omega)\phi_i(\boldsymbol{x}).$$

Using this and the Mercer's theorem, gives for any $\boldsymbol{x} \in D$,

$$\begin{aligned} \mathbf{Var}[a(\boldsymbol{x},\omega)] &= C(\boldsymbol{x},\boldsymbol{x}) = \sum_{i=1}^{\infty} \lambda_i \phi_i(\boldsymbol{x})^2 \\ \mathbf{Var}[a_J(\boldsymbol{x},\omega)] &= \widehat{C}(\boldsymbol{x},\boldsymbol{x}) = \sum_{i=1}^{J} \lambda_i \phi_i(\boldsymbol{x})^2 \end{aligned} \tag{1.23}$$

Hence

$$\mathbf{Var}[a(\boldsymbol{x},\omega)] - \mathbf{Var}[a_J(\boldsymbol{x},\omega)] = \sum_{i=J+1}^{\infty} \lambda_i \phi_i(\boldsymbol{x})^2 \geq 0 \to \mathbf{Var}[a(\boldsymbol{x},\omega)] \geq \mathbf{Var}[a_J(\boldsymbol{x},\omega)].$$

We always *underestimate* the variance.

1.9.3. A Practical Error Estimator

We could define

$$E_J := \frac{\int_D \mathbf{Var}[a(\boldsymbol{x},\omega) - a_J(\boldsymbol{x},\omega)]\mathrm{d}\boldsymbol{x}}{\int_D \mathbf{Var}[a(\boldsymbol{x},\omega)]\mathrm{d}\boldsymbol{x}} \tag{1.24}$$

If the RF has constant variance $\mathbf{Var}[a(\boldsymbol{x},\omega)] = \sigma^2$, then we have

$$E_J := \frac{\sigma^2|D| - \sum_{i=1}^{J} \lambda_i}{\sigma^2|D|,} \tag{1.25}$$

which is computable. If we know the eigenvalues, we can find J such that $E_J \leq \texttt{tol}$.

2. Numerical Methods for Generating RFs

Recall. The KL expansion of a random field $a : D \times \Omega \to \mathbf{R}$ is

$$a(\boldsymbol{x}, \omega) = \mu(\boldsymbol{x}) + \sum_{i=1}^{\infty} \sqrt{\lambda_i} \xi_i(\omega) \phi_i(\boldsymbol{x})$$

this is a function of $\boldsymbol{x} \in D$.

Key Point. When solving ODEs/PDEs with RFs inputs numerically, we often only need access to samples at a finite set of grid points $x_1, x_2, \ldots, x_N$. (see example 2.1)

***Example* 2.1.** Let $D = (0, 1)$, find $u : \overline{D} \to \mathbf{R}$ such that

$$-\frac{\mathrm{d}}{\mathrm{d}x}(a(x)\frac{\mathrm{d}u(x)}{\mathrm{d}x}) = 1 \quad x \in D.$$

with $u(0) = u(1) = 0$. We could approximate the solution using a *finite difference method.*

- Divide the interval $[0, 1]$ into N subintervals (or elements) of width $h = 1/N$ and label the vertices of these elements as $x_0, x_1, ..., x_N$. Then we have

$$-\frac{d}{dx}(a\left(x_i\right)\frac{du\left(x_i\right)}{dx}) = 1, \quad i = 1, \ldots, N-1 \tag{2.1}$$

  with $u(x_0) = u(x_N) = 0$.

- Using a centered finite difference scheme:

$$\frac{\mathrm{d}u}{\mathrm{d}x}(x_i) \approx \frac{u(x_i + \frac{h}{2}) - u(x_i - \frac{h}{2})}{h} = \frac{u(x_{i+1/2}) - u(x_{i-1/2})}{h} \tag{2.2}$$

  where $x_{i+1/2}$ is the midpoint of $[x_i, x_{i+1}]$ and similarly for $x_{i-1/2}$. Approximating the derivatives in (2.1) in this way and replacing $u\left(x_i\right)$ by U_i, where U_i denotes the resulting approximation to $u\left(x_i\right)$ gives a linear system of $N-1$ equations

$$-\left(\frac{a\left(x_{i-1/2}\right)}{h^2}\right)U_{i-1} + \left(\frac{a\left(x_{i-1/2}\right) + a\left(x_{i+1/2}\right)}{h^2}\right)U_i - \left(\frac{a\left(x_{i+1/2}\right)}{h^2}\right)U_{i+1} = 1 \tag{2.3}$$

  for $i = 1, \ldots, N-1$, together with $U_0 = 0$ and $U_N = 0$. If we consider a stochastic model where $a(x, \omega)$ is a stochastic process, then to perform a Monte Carlo simulation, we will need to generate samples of the random vector

$$\mathbf{a}(\omega) = \left[a\left(x_{1/2}, \omega\right), a\left(x_{3/2}, \omega\right), \ldots, , a\left(x_{N-(1/2)}, \omega\right)\right]^\top \tag{2.4}$$

  corresponding to $a(x, \omega)$ evaluated at the mid-points of the elements in the grid.

2.1. Generating Realisations of Gaussian RFs

Let $D \subset \mathbf{R}^d$ be a domain ($d = 1, 2$), and let $z(\boldsymbol{x}, \omega)$ be an associated second-order RF with mean function $\mu(\boldsymbol{x})$ and covariance function $C(\boldsymbol{x}, \boldsymbol{y})$.

2.1.1. Random fields to random vectors

Let $N \in \mathbf{N}$, for any $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N \in D$, we can define

$$\boldsymbol{Z}(\omega) := [z(\boldsymbol{x}_1, \omega), z(\boldsymbol{x}_2, \omega), \ldots, z(\boldsymbol{x}_N, \omega)]^\top \tag{2.5}$$

This is an $\mathbf{R}^N$-valued random variable $\boldsymbol{Z} : \Omega \to \mathbf{R}^N$.

- It has *mean vector*
$$\boldsymbol{\mu} := \mathbf{E}[\boldsymbol{Z}], \text{ where } [\mu]_i = \mu(\boldsymbol{x}_i), \quad i = 1, 2, \ldots, N.$$
- It has *Covariance matrix*
$$C_N := \mathbf{E}[(\boldsymbol{Z} - \boldsymbol{\mu})(\boldsymbol{Z} - \boldsymbol{\mu})^\top] \in \mathbf{R}^{N \times N},$$
where entries are $[C_N]_{ij} = C(\boldsymbol{x}_i, \boldsymbol{x}_j)$, $i, j = 1, \ldots, N$.

By theorem 1.20, the $N \times N$ covariance matrix C_N of $\boldsymbol{Z}$ is also symmetric and non-negative definite (or positive semidefinite) matrix, that is $\boldsymbol{v}^\top C_N \boldsymbol{v} \geq 0$, for all $\boldsymbol{v} \in \mathbf{R}^N$. From definition 1.22, we knwo that if $z(\boldsymbol{x}, \omega)$ is a Gaussian RF, then for any $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N$, $\boldsymbol{Z} \sim N(\boldsymbol{\mu}, C_N)$. That is, $\boldsymbol{Z}$ is always a *multivariate normal* random variable. Hence, if we need to generate realisations of a Gaussian RF at a set of grid points, then we only need to generate samples of random vectors $\boldsymbol{Z}$ with distribution $\boldsymbol{Z} \sim N(\boldsymbol{\mu}, C_N)$.

Standard approach is to use a *matrix fractorisation* of C_N. Suppose we can find an $N \times N$ matrix V such that
$$C_N = VV^\top.$$
We can create a sample of $\boldsymbol{Z} \sim N(\boldsymbol{\mu}, C_N)$ as follows:

Theorem 2.2 (Matrix Factorization Method)**.**
- *Generate a sample* $\boldsymbol{\xi} \sim N(\mathbf{0}, I_N)$*, where* $\boldsymbol{\xi} = [\xi_1, \xi_2, \ldots, \xi_N]^\top$*,* $\xi_i \sim N(0, 1)$ *iid.*
- *Factorize* $C_N = VV^\top$*.*
- *Form* $\boldsymbol{Z} = \boldsymbol{\mu} + V\boldsymbol{\xi}$*, and then* $\boldsymbol{Z} \sim N(\boldsymbol{\mu}, C_N)$*.*

2.2. Cholesky Factorization

Theorem 2.3 (Cholesky factorisation)**.** *If* C_N *is an* $N \times N$ *symmetric and positive definite matrix, i.e.,*
$$\boldsymbol{v}^\top C_N \boldsymbol{v} > 0 \quad \forall \boldsymbol{v} \in \mathbf{R}^N \setminus \{\mathbf{0}\},$$
then there exist an $N \times N$ *lower triangular* L *such that*
$$C_N = LL^\top. \tag{2.6}$$

Potential disadvantages for Cholesky method

- If C_N is not strictly positive definite, or has small eigenvalues very close to zero, then the method can fail.
- Since covariance matrices are usually dense, it can be very expensive:
  - Cost of Cholesky factorisation: $O(N^3)$,
  - Cost of Multiplication with L: $O(N^2)$.

  If we need M samples, then the cost is
$$O(N^3) + M \cdot O(N^2) \quad \text{where } N \text{ is the number of grid points} \tag{2.7}$$

2.3. Eigenvalue Decomposition

If C_N is real valued, symmetric and non-negative definite. We can factorise it as
$$C_N = U\Lambda U^\top$$
where

- Λ is the $N \times N$ diagonal matrix of eigenvalues, ordered such that
$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0.$$
- U is a matrix whose columns are eigenvectors
$$U = [\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_N] \in \mathbf{R}^{N \times N},$$
and we assume these are normalised such that
$$u_i^\top u_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad \Rightarrow U^\top U = I_N. \tag{2.8}$$

We could then manipulate the factorisation as
$$C_N = U\Lambda U^\top = U\Lambda^{1/2}\Lambda^{1/2}U^\top = U\Lambda^{1/2}(U\Lambda^{1/2})^\top \tag{2.9}$$
Here $\Lambda^{1/2}$ is the $N \times N$ diagonal matrix with diagonal entries,
$$\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_N}.$$
This is well-defined since C_N is non-negative definite. We can now use matrix factorisation method with $V = U\Lambda^{1/2}$, and this gives $\boldsymbol{Z} = \boldsymbol{\mu} + U\Lambda^{1/2}\boldsymbol{\xi}$ where $\boldsymbol{\xi} \sim N(\mathbf{0}, I_N)$.

2.4. Truncated Eigenvalue Decomposition

Note that
$$\boldsymbol{Z} = \boldsymbol{\mu} + U\Lambda^{1/2}\boldsymbol{\xi} = \boldsymbol{\mu} + \sum_{i=1}^{N} \sqrt{\lambda_i}\boldsymbol{u}_i\xi_i.$$

Recall. N is the number of grid points, and if we compare the cost with Cholesky method

- The cost of computing U and Λ is similar to Cholesky, $O(N^3)$.
- The cost of a multiplication with $V = U\Lambda^{1/2}$ is also $O(N^2)$.

However, We could make savings if we truncated the sum after $n < N$ terms.

Important. We assume that $\lambda_1 \leq \cdots \leq \lambda_N$. Let $n \in \mathbf{Z}$ with $n < N$ and define
$$\widehat{\boldsymbol{Z}} = \boldsymbol{\mu} + \sum_{i=1}^{n} \sqrt{\lambda_i}\boldsymbol{u}_i\xi_i, \quad \xi \sim N(0,1),\ iid. \tag{2.10}$$
We can also write
$$\widehat{\boldsymbol{Z}} = \boldsymbol{\mu} + U_n\Lambda_n^{1/2}\xi_n, \tag{2.11}$$
where $U_n = [\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n]$, $\Lambda_n = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, and $\boldsymbol{\xi}_n = [\xi_1, \ldots, \xi_n]^\top$.

Key Question. How good $\widehat{\boldsymbol{Z}}$ as an *approximation* to Z?

The random vector $\widehat{\boldsymbol{Z}}$ is Gaussian but the covariance matrix is not C_N. And it can be shown that
$$\widehat{\boldsymbol{Z}} \sim N(\boldsymbol{\mu}, \widehat{C}_n), \quad \text{where } \widehat{C}_n = U_n\Lambda_n U_n^\top \tag{2.12}$$
and we have the equality:
$$\frac{\|C_N - \widehat{C}_n\|_2}{\|C_N\|_2} = \frac{\lambda_{n+1}}{\lambda_1} \tag{2.13}$$
And also by considering the mean-square sense:
$$\mathbf{E}[\|\boldsymbol{Z} - \widehat{\boldsymbol{Z}}\|_2^2] = \sum_{i=n+1}^{N} \lambda_i. \tag{2.14}$$

2.5. Toeplitz and Circular Matrix

Definition 2.4. A *second order* RF is called *stationary* if the mean $\mu(\boldsymbol{x})$ is a constant and the covariance function

$$C(\boldsymbol{x}, \boldsymbol{y}) = c(\boldsymbol{x} - \boldsymbol{y}), \quad \boldsymbol{x}, \boldsymbol{y} \in D \tag{2.15}$$

for some function $c : D \to \mathbf{R}$ called the stationary covariance.

***Example* 2.5.** A mean zero RF with exponential covariance function

$$C(\boldsymbol{x}, \boldsymbol{y}) = \sigma^2 \exp\left(-\frac{\|\boldsymbol{x} - \boldsymbol{y}\|_2}{\ell}\right), \quad \boldsymbol{x}, \boldsymbol{y} \in D \subset \mathbf{R}^2 \tag{2.16}$$

is stationary since $C(\boldsymbol{x}, \boldsymbol{y}) = c(\boldsymbol{x} - \boldsymbol{y})$ where

$$c(\boldsymbol{x}) = \sigma^2 \exp\left(-\frac{\|\boldsymbol{x}\|_2}{\ell}\right) \tag{2.17}$$

Definition 2.6 (Toeplitz matrix)**.** An $N \times N$ real-valued matrix C is toeplitz if the (i, j)th entry

$$[C]_{i,j} = c_{i-j}$$

for some real numbers $c_{1-N}, \ldots, c_{N-1}$ In order words, the toeplitz matrix is constant along the diagonal and off diagonals:

$$C = \begin{pmatrix} c_0 & c_{-1} & \cdots & c_{2-N} & c_{1-N} \\ c_1 & c_0 & c_{-1} & \cdots & c_{2-N} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ c_{N-2} & \ddots & c_1 & c_0 & c_{-1} \\ c_{N-1} & c_{N-2} & \cdots & c_1 & c_0 \end{pmatrix} \tag{2.18}$$

We only need to store the first column and row to reconstruct the whole toeplitz matrix.

NOTE. A *symmetric* toeplitz matrix is fully defined by its first column.

$$\boldsymbol{c}_1 = [c_0, c_1, \ldots, c_{N-1}]^\top \in \mathbf{R}^N.$$

***Example* 2.7.** Let $D = [0, 1]$, and let $z(x, \omega)$ be a mean-zero Gaussian RF with stationary covariance

$$c(x) = e^{-|x|/\ell}. \tag{2.19}$$

Divide D into 100 uniform intervals and choose the grid points

$$x_i = \frac{i-1}{100}, \quad i = 1, 2, \ldots, 101.$$

The plot of covariance matrix is shown in figure 5.

Definition 2.8 (Circulant matrix)**.** An $N \times N$ real-valued Topelitz matrix C is circulant if each column is a circular (clockwise) shift of the preceding (previous) column.

$$C = \begin{pmatrix} c_0 & c_{N-1} & \cdots & c_2 & c_1 \\ c_1 & c_0 & c_{N-1} & \cdots & c_2 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ c_{N-2} & \ddots & c_1 & c_0 & c_{N-1} \\ c_{N-1} & c_{N-2} & \cdots & c_1 & c_0 \end{pmatrix} \tag{2.20}$$

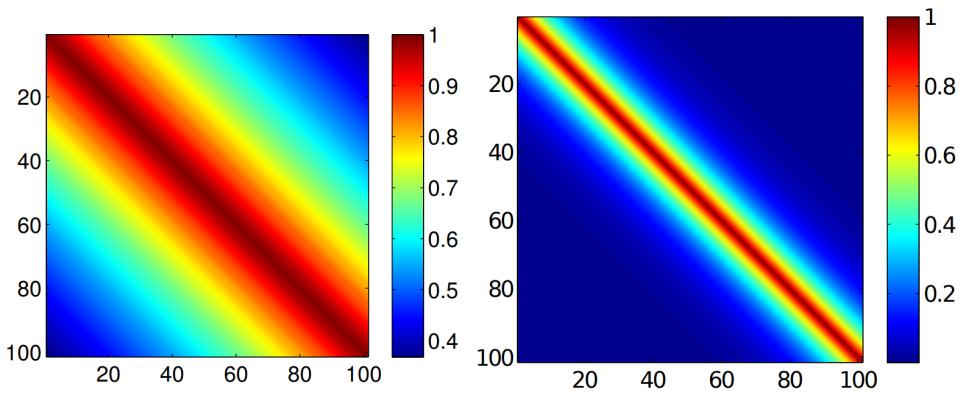

FIGURE 5: MATLAB imagesc plots of the covariance matrix C_N associated with a mean zero Gaussian process with exponential covariance function with $\ell = 1$ (left) and $\ell = 1/10$ (right), sampled at $N = 101$ equally spaced points in $D = [0, 1]$.

***Example* 2.9.** Example of Circulant matrix:

$$C = \begin{pmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$$

is circulant.

Definition 2.10 (Hermitian vector)**.** A vector $\boldsymbol{v} = [v_0, v_1, \ldots, v_{N-1}]^\top \in \mathbf{C}^N$ is said to be Hermitian if

$$v_0 \in \mathbf{R}, \quad \text{and } v_j = \overline{v_{N-j}}, \quad j = 1, 2, \ldots, N-1. \tag{2.21}$$

Remark 2.11. Again, circulant matrix is uniquely determined by the first column

$$\boldsymbol{c}_1 = [c_0, c_1, \ldots, c_{N-1}]^\top \in \mathbf{R}^N.$$

If the matrix is also symmetric, the first column has a special structure.

$$c_i = c_j \;\text{ if }\; i + j = N, \text{ for } i, j = 1, 2, \ldots, N-1. \tag{2.22}$$

i.e., the first column is real Hermitian vector.

Theorem 2.12 (Topelitz Covariance Matrix)**.** *Let $D = [a, b]$. If*

- *$z(x, \omega)$ be a stationary stochastic process on D with* stationary *covariance function $c(x)$.*
- *Suppose we evaluate $z(x, \omega)$ at N* uniformly spaced *points $x_1, \ldots, x_N$.*

Then the associated covariance matrix C_N is always Toeplitz.

Proof. (Note the points are labelled $x_1, \ldots, x_N$ here). Since the points are uniformly spaced on $[a, b]$, we have $x_i = a + (i-1)h, i = 1, 2, \ldots, N$, where $h = (b-a)/(N-1)$. (If there are N points, there are $N-1$ intervals). Then $x_i - x_j = (i-j)h$. The $(i, j)th$ entry of the covariance matrix is

$$[C_N]_{i,j} = \underbrace{C(x_i, x_j) = c(x_i - x_j)}_{\text{Stationary covariance function}} = c((i-j)h)$$

where $c(x)$ is the stationary covariance function. Hence $[C_N]_{i,j}$ is constant for all pairs of point x_i, x_j for which $(i-j)$ is constant. This means that C_N is Toeplitz (by definition 2.6), since all entries along a fixed diagonal have the same value of $i - j$. □

2.6. Discrete Fourier Transform

Definition 2.13 (Discrete Fourier Transform (DFT))**.** The DFT of a vector $\boldsymbol{v} \in \mathbf{C}^N$ is the vector $\widehat{\boldsymbol{v}} \in \mathbf{C}^N$ with entries

$$\widehat{v}_k := \sum_{j=1}^{N} v_j \omega^{(k-1)(j-1)}, \quad k = 1, 2, \ldots, N. \tag{2.23}$$

where $\omega := e^{-2\pi i/N}$. (This is the Nth unity, and distinguish ω with the iid. random normal distribution samples)

Equivalently, we could write the definition as

$$\widehat{\boldsymbol{v}} := \sqrt{N} W \boldsymbol{v} \in \mathbf{C}^N, \tag{2.24}$$

where $W \in \mathbf{C}^{N\times N}$ is the *Fourier matrix* has entries

$$[W]_{k,j} = \frac{1}{\sqrt{N}} \omega^{(k-1)(j-1)}, \quad k, j = 1, \ldots, N. \tag{2.25}$$

NOTE.

- The MATLAB command `fft(v)` will perform a DFT on the vector $\boldsymbol{v}$.
- This important algorithm will reduce the cost of $(\mathbf{C}^{N\times N} \times \mathbf{C}^N) = O(N^2)$ to $O(N \log N)$.

Definition 2.14 (Inverse Discrete Fourier Transform (IDFT))**.** The IDFT of $\widehat{\boldsymbol{v}} \in C^N$ is the vector

$$\boldsymbol{v} := \frac{1}{\sqrt{N}} W^* \widehat{\boldsymbol{v}} \in \mathbf{C}^N \tag{2.26}$$

where W^* is the *Hermitian transpose* of W. That is, W^* satisfies

$$[W^*]_{k,j} = \overline{W_{j,k}}, \quad k, j = 1, 2, \ldots, N.$$

NOTE.

- The MATLAB command `ifft(v)` will perform a IDFT on the vector $\boldsymbol{v}$.
- This important algorithm will reduce the cost of $(\mathbf{C}^{N\times N} \times \mathbf{C}^N) = O(N^2)$ to $O(N \log N)$.
- The DFT and IDFT of a *Hermitian vector* is always *real-valued*.

2.6.1. Factorization of Circulant matrix

Theorem 2.15. *Let $C \in \mathbf{R}^{N\times N}$ be a circulant matrix with first column $\boldsymbol{c}_1$, then*

$$C = W \Lambda W^* \tag{2.27}$$

- *$W \in \mathbf{C}^{N\times N}$ is the $N \times N$ Fourier matrix.*
- *Λ is the $N \times N$ diagonal matrix of eigenvalues with diagonal entries*

$$\boldsymbol{d} = \sqrt{N} W^* \boldsymbol{c}_1. \tag{2.28}$$

- We can compute the eigenvalues using the IDFT of $\boldsymbol{c}_1$ in MATLAB `d=N*ifft(c1)`.
- If C is symmetric, and circulant, by remark 2.11, then $\boldsymbol{c}_1$ is a Hermitian vector, hence $\boldsymbol{d} \in \mathbf{R}^N$, i.e., the eigenvalues are all real.

- If C is a valid covariance matrix as well as circulant, i.e., the matrix is *symmetric, non-negative definite (positive semi-definite), and circulant*, then by theorem 2.15 we have

$$C = W\Lambda W^* = W\Lambda^{1/2}\Lambda^{1/2}W^* = (W\Lambda^{1/2})(W\Lambda^{1/2})^* \tag{2.29}$$

Problem. Suppose C_N is $N \times N$ covariance matrix that is also circulant, then theorem 2.15 and equation (2.29) give

$$C_N = VV^*, \quad \text{where } V := W\Lambda^{1/2}. \tag{2.30}$$

If we form $\boldsymbol{Z} = \boldsymbol{\mu} + W\Lambda^{1/2}\boldsymbol{\xi}$, where $\boldsymbol{\xi} \sim N(\mathbf{0}, I_N)$, what's the distribution of $\boldsymbol{Z}$?

$$\begin{aligned}
\mathbf{E}[\boldsymbol{Z}] &= \boldsymbol{\mu}, \\
\operatorname{Cov}[\boldsymbol{Z}] &= \operatorname{Cov}[(\boldsymbol{Z}-\boldsymbol{\mu})(\boldsymbol{Z}-\boldsymbol{\mu})^\top] \\
&= \operatorname{Cov}[W\Lambda^{1/2}\boldsymbol{\xi}\boldsymbol{\xi}^\top\Lambda^{1/2}W^\top] \\
&= W\Lambda^{1/2}\operatorname{Cov}[\boldsymbol{\xi}\boldsymbol{\xi}^\top]\Lambda^{1/2}W^\top \\
&= W\Lambda^{1/2}I_N\Lambda^{1/2}W^\top = W\Lambda W^\top \neq W\Lambda W^*.
\end{aligned} \tag{2.31}$$

2.7. Complex-Valued Random Variables

Given 2 $\mathbf{R}^N$-valued random variables

$$\boldsymbol{X}_1, \boldsymbol{X}_2 : \Omega \to \mathbf{R}^N. \tag{2.32}$$

We can always create a $\mathbf{C}^N$-valued random variable via

$$\boldsymbol{X} = \boldsymbol{X}_1 + i\boldsymbol{X}_2 \tag{2.33}$$

where $\boldsymbol{X}_1 = \mathcal{R}\mathrm{e}(\boldsymbol{X})$, $\boldsymbol{X}_2 = \mathcal{I}\mathrm{m}(\boldsymbol{X})$.

Definition 2.16 (Complex R.V.s)**.** If $\boldsymbol{X} = [X_1, X_2, \ldots, X_N]^\top$ is a $\mathbf{C}^N$-valued r.v., then

- The *mean vector* $\boldsymbol{\mu} \in \mathbf{C}^N$ where

$$[\boldsymbol{\mu}]_j = \mathbf{E}[X_j], \quad j = 1, 2, \ldots, N. \tag{2.34}$$

- The *covariance matrix* $C_{\boldsymbol{X}} \in \mathbf{C}^{N\times N}$ where

$$C_{\boldsymbol{X}} = \mathbf{E}[(\boldsymbol{X}-\boldsymbol{\mu})(\boldsymbol{X}-\boldsymbol{\mu})^*] \in \mathbf{C}^{N\times N} \tag{2.35}$$

  where $*$ denotes the Hermitian transpose. That is,

$$[C_{\boldsymbol{X}}]_{k,j} = \operatorname{Cov}(X_k, X_j) = \mathbf{E}[(X_k - \mathbf{E}[X_k])\overline{(X_j - \mathbf{E}[X_j])}]. \tag{2.36}$$

***Example* 2.17.** Let $\boldsymbol{X}_1$, $\boldsymbol{X}_2$ be $\mathbf{R}^N$ random variables with *mean zero*. Define $\boldsymbol{X} = \boldsymbol{X}_1 + i\boldsymbol{X}_2$. Then

$$\begin{aligned}
\mathbf{E}[\boldsymbol{X}] &= \mathbf{E}[\boldsymbol{X}_1] + i\,\mathbf{E}[\boldsymbol{X}_2] = \mathbf{0} \\
C_{\boldsymbol{X}} &= \mathbf{E}[(\boldsymbol{X} - \mathbf{E}[\boldsymbol{X}])(\boldsymbol{X} - \mathbf{E}[\boldsymbol{X}])^*] = \mathbf{E}[\boldsymbol{X}\boldsymbol{X}^*]
\end{aligned} \tag{2.37}$$

Theorem 2.18 (Uncorrelated $\to$ Real-valued)**.** *Suppose we have a complex valued random variable $\boldsymbol{X}$. If $\mathcal{R}\mathrm{e}(\boldsymbol{X})$ and $\mathcal{I}\mathrm{m}(\boldsymbol{X})$ are uncorrelated and mean zero, then the corresponding covariance $C_{\boldsymbol{X}}$ is $\mathbf{R}$-valued.*

Proof. For complex valued random variable $\boldsymbol{X} = \boldsymbol{X}_1 + i\boldsymbol{X}_2$, the covariance matrix is

$$\begin{aligned} C_{\boldsymbol{X}} &= \mathbf{E}\left[(\boldsymbol{X}_1 + i\boldsymbol{X}_2)(\boldsymbol{X}_1 + i\boldsymbol{X}_2)^*\right] = \mathbf{E}\left[(\boldsymbol{X}_1 + i\boldsymbol{X}_2)(\boldsymbol{X}_1 - i\boldsymbol{X}_2)^\top\right] \\ &= \mathbf{E}\left[\boldsymbol{X}_1\boldsymbol{X}_1^\top + \boldsymbol{X}_2\boldsymbol{X}_2^\top\right] + i\,\mathbf{E}\left[-\boldsymbol{X}_1\boldsymbol{X}_2^\top + \boldsymbol{X}_2\boldsymbol{X}_1^\top\right] \end{aligned}$$

Since $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ are assumed to be uncorrelated, $\mathrm{Cov}(\boldsymbol{X}_1, \boldsymbol{X}_2) = \mathbf{E}\left[\boldsymbol{X}_1\boldsymbol{X}_2^\top\right] = \mathbf{0} = \mathbf{E}\left[\boldsymbol{X}_2\boldsymbol{X}_1^\top\right]$. The second term is zero and we have

$$C_{\boldsymbol{X}} = \mathbf{E}\left[\boldsymbol{X}_1\boldsymbol{X}_1^\top + \boldsymbol{X}_2\boldsymbol{X}_2^\top\right] = \mathbf{E}[\boldsymbol{X}_1\boldsymbol{X}_1^\top] + \mathbf{E}[\boldsymbol{X}_2\boldsymbol{X}_2^\top] = \mathrm{Cov}(\boldsymbol{X}_1) + \mathrm{Cov}(\boldsymbol{X}_2),$$

which is the sum of the covariance matrices of $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ and, in particular, is real-valued. □

Lemma 2.19 (Real-valued → Uncorrelated)**.** *Let $\boldsymbol{X} = \boldsymbol{X}_1 + i\boldsymbol{X}_2$ where $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ are $\mathbf{R}^N$-valued random variables with mean zero.*

- *If $\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^*]$ and $\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top]$ are real-valued, then $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ are uncorrelated.*
- *If in addition, $\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \mathbf{0}$, then the covariance matrices of $\boldsymbol{X}_1$, and $\boldsymbol{X}_2$ satisfy*

$$C_{\boldsymbol{X}_1} = C_{\boldsymbol{X}_2} = \frac{\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^*]}{2} = \frac{C_{\boldsymbol{X}}}{2}. \tag{2.38}$$

Proof. Let $\boldsymbol{X} = \boldsymbol{X}_1 + i\boldsymbol{X}_2$, and we have $\overline{\boldsymbol{X}} = \boldsymbol{X}_1 - i\boldsymbol{X}_2$. Hence we could rewrite

$$\boldsymbol{X}_1 = \frac{\boldsymbol{X} + \overline{\boldsymbol{X}}}{2}, \quad \boldsymbol{X}_2 = \frac{\boldsymbol{X} - \overline{\boldsymbol{X}}}{2i} \tag{2.39}$$

then we could evaluate

$$\begin{aligned} \mathrm{Cov}(\boldsymbol{X}_1, \boldsymbol{X}_2) &= \mathbf{E}[\boldsymbol{X}_1\boldsymbol{X}_2^\top] \\ &= \frac{1}{4i}\mathbf{E}[(\boldsymbol{X} + \overline{\boldsymbol{X}})(\boldsymbol{X} - \overline{\boldsymbol{X}})^\top] \\ &= \frac{1}{4i}\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top - \boldsymbol{X}\boldsymbol{X}^* + \overline{\boldsymbol{X}}\boldsymbol{X}^\top - \overline{\boldsymbol{X}}\boldsymbol{X}^*] \\ &= \frac{1}{4i}(\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top] - \mathbf{E}[\boldsymbol{X}\boldsymbol{X}^*] + \mathbf{E}[\overline{\boldsymbol{X}}\boldsymbol{X}^\top] - \mathbf{E}[\overline{\boldsymbol{X}}\boldsymbol{X}^*]) \end{aligned} \tag{2.40}$$

By assumptions that $\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^*]$ and $\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top]$ are real valued, then we could have

$$\begin{aligned} \mathbf{E}[\boldsymbol{X}\boldsymbol{X}^*] &= \overline{\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^*]} = \mathbf{E}[\overline{\boldsymbol{X}}\boldsymbol{X}^\top], \\ \mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top] &= \overline{\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top]} = \mathbf{E}[\overline{\boldsymbol{X}}\boldsymbol{X}^*]. \end{aligned} \tag{2.41}$$

then we have the cancellation, and we have $\mathrm{Cov}(\boldsymbol{X}_1, \boldsymbol{X}_2) = \mathbf{0}$ as required.

$$\begin{aligned} C_{\boldsymbol{X}_1} = \mathbf{E}[\boldsymbol{X}_1\boldsymbol{X}_1^\top] &= \frac{1}{4}\mathbf{E}[(\boldsymbol{X} + \overline{\boldsymbol{X}})(\boldsymbol{X} + \overline{\boldsymbol{X}})^\top] \\ &= \frac{1}{4}\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top + \boldsymbol{X}\boldsymbol{X}^* + \overline{\boldsymbol{X}}\boldsymbol{X}^\top + \overline{\boldsymbol{X}}\boldsymbol{X}^*] \\ &= \frac{1}{2}\mathbf{E}[\boldsymbol{X}\boldsymbol{X}^*] = \frac{C_{\boldsymbol{X}}}{2}. \quad \text{By } \mathbf{E}[\boldsymbol{X}\boldsymbol{X}^\top] = \mathbf{E}[\overline{\boldsymbol{X}}\boldsymbol{X}^*] = 0 \text{ and equation (2.41)} \end{aligned} \tag{2.42}$$

□

Definition 2.20 (Complex Gaussian Distribution)**.** An $\mathbf{C}^N$ random variable $\boldsymbol{X} = \boldsymbol{X}_1 + i\boldsymbol{X}_2$ ($\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ are real valued random variables) follows the complex Gaussian distribution

$$\boldsymbol{X} \sim CN(\mathbf{0}, C_{\boldsymbol{X}}) \tag{2.43}$$

with real-valued covariance matrix $C_{\boldsymbol{X}} \in \mathbf{R}^{N\times N}$ if

- $\boldsymbol{X}_1$ and $\boldsymbol{X}_2$ are independent
- $\boldsymbol{X}_1, \boldsymbol{X}_2 \sim N\left(\mathbf{0}, \frac{C_X}{2}\right)$.

***Example* 2.21.** Let $\boldsymbol{X}_1, \boldsymbol{X}_2 \sim N(\mathbf{0}, I_N)$ be independent and form $\boldsymbol{\xi} = \boldsymbol{X}_1 + i\boldsymbol{X}_2$, then we have

$$\boldsymbol{\xi} \sim CN(\mathbf{0}, 2I_N). \tag{2.44}$$

2.8. Circulant Covariance Matrix

Suppose C_N is an $N \times N$ real-valued covariance matrix that is

- a valid covariance matrix
- circulant

Recall (Usual factorisation method (based on theorem 2.2))**.**

- Generate a sample of $\boldsymbol{\xi} \sim N(\mathbf{0}, I_N)$.
- Compute Λ (the eigenvalues of C_N).
- Form $\boldsymbol{Z} = W\Lambda^{1/2}\boldsymbol{\xi}$.

We will introduce a new method for generating independent samples from $N(\mathbf{0}, C_N)$. However, from equation (2.31), we recognize that the method we introduced in theorem 2.2 will not gives the desired distribution $N(\mathbf{0}, C_N)$. In particular, the resulting $\boldsymbol{Z}$ is complex-valued.

2.8.1. Modified Factorisation Method

Corollary 2.22. *For circulant symmetric and non-negative definite matrix C_N, we could modify the theorem 2.2*

1. *Generate a sample of $\boldsymbol{\xi} \sim CN(\mathbf{0}, 2I_N)$.*
2. *Compute Λ (the eigenvalues of C_N).*
3. *Form $\boldsymbol{Z} = W\Lambda^{1/2}\boldsymbol{\xi}$.*

Then $\boldsymbol{Z} \sim CN(\mathbf{0}, 2C_N)$.

Since C_N is circulant, hence step (2) can be done using IDFT on $\boldsymbol{c}_1$, and step (3) can be done using DFT. Hence the resulting cost will be $O(N \log N)$.

Theorem 2.23. *Using the method in corollary 2.22, we have*

$$\mathcal{R}\mathrm{e}(\boldsymbol{Z}), \mathcal{I}\mathrm{m}(\boldsymbol{Z}) \sim N(\mathbf{0}, C_N) \quad \textit{and independent.} \tag{2.45}$$

Proof. It is easy to see that $\boldsymbol{Z}$ is a linear transformation of a multivariate Gaussian random variable $\boldsymbol{\xi}$, hence $\boldsymbol{Z}$ is a multivariate Gaussian random variable.

$$\mathbf{E}[\boldsymbol{Z}] = W\Lambda^{1/2}\,\mathbf{E}[\boldsymbol{\xi}] = \mathbf{0}. \tag{2.46}$$

Hence if we define $\boldsymbol{Z} = \mathcal{R}\mathrm{e}(\boldsymbol{Z}) + i\,\mathcal{I}\mathrm{m}(\boldsymbol{Z}) = \boldsymbol{Z}_1 + i\boldsymbol{Z}_2$, then $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_2$ both mean $\mathbf{0}$.

$$\begin{aligned}\mathbf{E}[\boldsymbol{Z}\boldsymbol{Z}^*] &= \mathbf{E}[W\Lambda^{1/2}\boldsymbol{\xi}\boldsymbol{\xi}^*\Lambda^{1/2}W^*] \\ &= W\Lambda^{1/2}\,\mathbf{E}[\boldsymbol{\xi}\boldsymbol{\xi}^*]\Lambda^{1/2}W^* \\ &= 2W\Lambda W^* = 2C_N \in \mathbf{R}^N\end{aligned} \tag{2.47}$$

Since $\boldsymbol{\xi} \sim CN(\mathbf{0}, 2I_N)$, hence by definition 2.20 if we write $\boldsymbol{\xi} = \boldsymbol{\xi}_1 + i\boldsymbol{\xi}_2$, then $\boldsymbol{\xi}_1$ and $\boldsymbol{\xi}_2$ are independent ($\mathbf{E}[\boldsymbol{\xi}_1\boldsymbol{\xi}_2^\top] = \mathbf{E}[\boldsymbol{\xi}_2\boldsymbol{\xi}_1^\top] = \mathbf{0}$) and distributed as $N(\mathbf{0}, I_N)$.

$$\begin{aligned}
\mathbf{E}[\boldsymbol{Z}\boldsymbol{Z}^\top] &= \mathbf{E}[W\Lambda^{1/2}\boldsymbol{\xi}\boldsymbol{\xi}^\top\Lambda^{1/2}W^\top] \\
&= W\Lambda^{1/2}\,\mathbf{E}[\boldsymbol{\xi}\boldsymbol{\xi}^\top]\Lambda^{1/2}W^\top \\
\mathbf{E}[\boldsymbol{\xi}\boldsymbol{\xi}^\top] &= \mathbf{E}[(\boldsymbol{\xi}_1 + i\boldsymbol{\xi}_2)(\boldsymbol{\xi}_1 + i\boldsymbol{\xi}_2)^\top] \\
&= \mathbf{E}[\boldsymbol{\xi}_1\boldsymbol{\xi}_1^\top] + i\,\mathbf{E}[\boldsymbol{\xi}_1\boldsymbol{\xi}_2^\top] + i\,\mathbf{E}[\boldsymbol{\xi}_2\boldsymbol{\xi}_1^\top] - \mathbf{E}[\boldsymbol{\xi}_2\boldsymbol{\xi}_2^\top] = \mathbf{0} \\
\mathbf{E}[\boldsymbol{Z}\boldsymbol{Z}^\top] &= \mathbf{0} \in \mathbf{R}^N.
\end{aligned} \tag{2.48}$$

By lemma 2.19, $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_2$ are mean zero, $\mathbf{E}[\boldsymbol{Z}\boldsymbol{Z}^*], \mathbf{E}[\boldsymbol{Z}\boldsymbol{Z}^\top] \in \mathbf{R}^N$, the we could conclude that

$$\boldsymbol{Z}_1, \boldsymbol{Z}_2 \sim N(\mathbf{0}, C_N), \quad \text{and independent.} \tag{2.49}$$

□

2.9. Circulant Embedding Method

Recall.

- We now have the method for generating a pair of samples from $N(\mathbf{0}, C_N)$ with C_N is c irculant (from corollary 2.22 and theorem 2.23).
- Unfortunately, covariance matrices are *not* usually c irculant.

Let $D \subset \mathbf{R}$ and let $z(x, \omega)$ be a mean-zero Gaussian RF, if

- the sample points $x_1, x_2, \ldots, x_N$ are uniformly spaced.
- the covariance function is stationary: $C(x, y) = c(x - y)$.

Then $\boldsymbol{Z} = [z(x_1, \omega), \ldots, z(x_N, \omega)]^\top \sim N(\mathbf{0}, C_N)$ where C_N is always T oeplitz (theorem 2.12).

2.9.1. Circulant Embedding

Any s ymmetric Toeplitz matrix C_N can be embedded inside a *larger* s ymmetric circulant matrix $\widetilde{C}$ of the form

$$\widetilde{C} = \left(\begin{array}{c|c} C_N & A^\top \\ \hline A & B \end{array}\right) \tag{2.50}$$

***Example* 2.24** ($N = 4$)**.** For

$$C_N = \begin{pmatrix} 5 & 2 & 3 & 4 \\ 2 & 5 & 2 & 3 \\ 3 & 2 & 5 & 2 \\ 4 & 3 & 2 & 5 \end{pmatrix} \tag{2.51}$$

Clearly, this matrix is symmetric and Toeplitz. This can be embedded into a larger s ymmetric circulant matrix.

$$\tilde{C} = \left(\begin{array}{cccc|cc} 5 & 2 & 3 & 4 & 3 & 2 \\ 2 & 5 & 2 & 3 & 4 & 3 \\ 3 & 2 & 5 & 2 & 3 & 4 \\ 4 & 3 & 2 & 5 & 2 & 3 \\ \hline 3 & 4 & 3 & 2 & 5 & 2 \\ 2 & 3 & 4 & 3 & 2 & 5 \end{array}\right) \tag{2.52}$$

If we focus on the first column of $\widetilde{C}$, $\widetilde{\boldsymbol{c}_1} = [5, 2, 3, 4, 3, 2]^\top$ which is a Hermitian vector.

2.9.2. Minimal circulant extension

Definition 2.25. Let C_N be an $N \times N$ symmetric Toeplitz matrix with first column

$$\boldsymbol{c}_1 = [c_0, c_1, \ldots, c_{N-2}, c_{N-1}]^\top. \tag{2.53}$$

The *minimal circulant extension* $\widetilde{C}$ is the symmetric and circulant matrix of size $P \times P$ where $P = 2N - 2$ with first column

$$\widetilde{\boldsymbol{c}_1} = [c_0, \underbrace{c_1, \ldots, c_{N-2}}_{\substack{\text{take this part,}\\ \text{flip the order}}}, c_{N-1} | c_{N-2}, \ldots, c_1]^\top. \tag{2.54}$$

That is,

$$\widetilde{C} = \left(\begin{array}{c|c} C_N & A^\top \\ \hline A & B \end{array}\right) \tag{2.55}$$

where $A \in \mathbf{R}^{(N-2)\times N}$, and $B \in \mathbf{R}^{(N-2)\times(N-2)}$.

2.9.3. Recovering Samples

If the extension matrix is a *valid covariance matrix*, then we can generate samples from

$$\widetilde{\boldsymbol{Z}} \sim N(\mathbf{0}, \widetilde{C}) \quad \widetilde{C} \text{ from equation (2.55)}$$

using the DFT-method. Such samples have the structure

$$\widetilde{\boldsymbol{Z}} = \left(\frac{\boldsymbol{Z}}{\boldsymbol{Y}}\right) \quad \boldsymbol{Z} \in \mathbf{R}^N,\ \boldsymbol{Y} \in \mathbf{R}^{N-2}$$

and it can be shown that $\boldsymbol{Z} \sim N(\mathbf{0}, C_N)$.

Proof. Write $\widetilde{\boldsymbol{Z}} = [\boldsymbol{Z}, \boldsymbol{Y}]^\top$ where $\boldsymbol{Z}$ is the $\mathbf{R}^N$-valued random variable of interest. If $\widetilde{\boldsymbol{Z}} \sim N(\mathbf{0}, \widetilde{C})$ then clearly $\boldsymbol{Z}$ has mean vector zero. The covariance matrix of $\boldsymbol{Z}$ is $\mathbf{E}\left[\boldsymbol{Z}\boldsymbol{Z}^\top\right]$ and the covariance matrix of $\widetilde{\boldsymbol{Z}}$ is $\widetilde{C} = \mathbf{E}\left[\widetilde{\boldsymbol{Z}}\widetilde{\boldsymbol{Z}}^\top\right]$. We have

$$\widetilde{C} = \begin{pmatrix} \mathbf{E}[\boldsymbol{Z}\boldsymbol{Z}^\top] & \mathbf{E}[\boldsymbol{Z}\boldsymbol{Y}^\top] \\ \mathbf{E}[\boldsymbol{Y}\boldsymbol{Z}^\top] & \mathbf{E}[\boldsymbol{Y}\boldsymbol{Y}^\top] \end{pmatrix} \underbrace{=}_{\text{equation (2.55)}} \begin{pmatrix} C_N & B^\top \\ B & D \end{pmatrix} \tag{2.56}$$

Hence the covariance matrix of $\boldsymbol{Z}$, $\mathbf{E}[\boldsymbol{Z}\boldsymbol{Z}^\top]$, is C_N. Finally, $\boldsymbol{Z}$ is Gaussian because it is a linear transformation of the Gaussian random variable $\widetilde{\boldsymbol{Z}}$. □

Theorem 2.26 (Circulant Embedding Method)**.** *To draw a pair of independent samples from $N(\mathbf{0}, C_N)$ when C_N is Toeplitz:*

- *Construct first column $\boldsymbol{c}_1$ of C_N*
- *Create first column $\widetilde{\boldsymbol{c}}_1$ of the minimal circulant extension $\widetilde{C}$*
- *Obtain two independent samples from $N(\mathbf{0}, \widetilde{C})$ (apply IDFT & DFT from corollary 2.22, and using theorem 2.23)*
- *Extract two independent samples from $N(\mathbf{0}, C_N)$*

2.10. Approximate Circulant Embedding

Recall. Any s ymmetric Toeplitz matrix C_N can be embedded inside a *larger* s ymmetric circulant matrix $\widetilde{C}$ of the form

$$\widetilde{C} = \left(\begin{array}{c|c} C_N & A^\top \\ \hline A & B \end{array}\right) \tag{2.57}$$

If $\widetilde{C}$ is *non-negative definite*, we can generate samples from $N(\mathbf{0}, \widetilde{C})$ and easily recover samples from $N(\mathbf{0}, C_N)$ using theorem 2.26.

However, the method is not working if we have a invalid (have negative eigenvalues) $\widetilde{C}$.

2.10.1. Matrix Splitting

If we have

$$\widetilde{C} = W\Lambda W^*.$$

We can split Λ into two order diagonal matrices,

$$\Lambda_+ - \Lambda_- \tag{2.58}$$

Let $\boldsymbol{d}$ be the vector containing the eigenvalues of $\widetilde{C}$ and define

$$[\Lambda_+]_{i,i} = \begin{cases} d_i & \text{if } d_i \geq 0 \\ 0 & \text{otherwise} \end{cases}, \qquad [\Lambda_-]_{i,i} = \begin{cases} -d_i & \text{if } d_i < 0 \\ 0 & \text{otherwise} \end{cases} \tag{2.59}$$

2.10.2. Approximate Circulant Embedding

We now have the *matrix splitting*:

$$\widetilde{C} = W\Lambda_+ W^* - W\Lambda_- W^* := \widetilde{C}_{pos} - \widetilde{C}_{neg} \approx \widetilde{C}_{pos}, \tag{2.60}$$

if the negative eigenvalues are small.

- The matrix $\widetilde{C}_{pos}$ is circulant and *non-negative definite*.
- We can apply the method (corollary 2.22) with $\widetilde{C}_{pos}$ in place of $\widetilde{C}$:

$$\widehat{\boldsymbol{Z}} = W\Lambda_+^{1/2}\boldsymbol{\xi}, \quad \text{where } \boldsymbol{\xi} \sim CN(\mathbf{0}, 2I_P). \tag{2.61}$$

  It can be shown that

$$\widehat{\boldsymbol{Z}} \sim CN(\mathbf{0}, 2\widetilde{C}_{pos}), \qquad \mathcal{R}\mathrm{e}(\widehat{\boldsymbol{Z}}), \mathcal{I}\mathrm{m}(\widehat{\boldsymbol{Z}}) \sim N(\mathbf{0}, \widetilde{C}_{pos}) \tag{2.62}$$

  Proof. $\mathcal{R}\mathrm{e}(\widehat{\boldsymbol{Z}})$ and $\mathcal{I}\mathrm{m}(\widehat{\boldsymbol{Z}})$ are both linear transformation of Gaussian $\mathcal{R}\mathrm{e}(\boldsymbol{\xi})$ and $\mathcal{I}\mathrm{m}(\boldsymbol{\xi})$. Hence $\mathcal{R}\mathrm{e}(\widehat{\boldsymbol{Z}})$ and $\mathcal{I}\mathrm{m}(\widehat{\boldsymbol{Z}})$ are Gaussian.

$$\begin{aligned} \mathbf{E}[\widehat{\boldsymbol{Z}}\widehat{\boldsymbol{Z}}^*] &= \mathbf{E}[W\Lambda_+^{1/2}\boldsymbol{\xi}\boldsymbol{\xi}^*\Lambda_+^{1/2}W^*] \\ &= W\Lambda_+^{1/2}(2I_P)\Lambda_+^{1/2}W^* \\ &= 2W\Lambda_+W^* = 2\widetilde{C}_{pos} \in \mathbf{R}^{P\times P} \\ \mathbf{E}[\widehat{\boldsymbol{Z}}\widehat{\boldsymbol{Z}}] &= \mathbf{0} \quad \text{By proof of theorem 2.23} \end{aligned} \tag{2.63}$$

  By lemma 2.19, $\mathcal{R}\mathrm{e}(\widehat{\boldsymbol{Z}})$ and $\mathcal{I}\mathrm{m}(\widehat{\boldsymbol{Z}})$ are independent and

$$\mathcal{R}\mathrm{e}(\widehat{\boldsymbol{Z}}), \mathcal{I}\mathrm{m}(\widehat{\boldsymbol{Z}}) \sim N(\mathbf{0}, \widetilde{C}_{pos})$$

□

2.10.3. Recovering Approximate Samples

Let $\boldsymbol{Z}_1$ and $\boldsymbol{Z}_2$ denote the first N components of these random vectors in equation (2.62). Then

$$\boldsymbol{Z}_1, \boldsymbol{Z}_2 \sim N(\mathbf{0}, \widehat{C}) \tag{2.64}$$

where $\widehat{C}$ is the $N \times N$ leading block of $\widetilde{C}_{pos}$.

$$\widetilde{C}_{pos} = \left(\begin{array}{c|c} \widehat{C} & A^\top \\ \hline A & B \end{array}\right) \tag{2.65}$$

However, since $\widehat{C} \neq C_N$, we do note have samples from the target distribution $N(\mathbf{0}, C_N)$.

Theorem 2.27. *It can be shown that,*

$$\|C_N - \widehat{C}\|_2 \leq \rho\left(\Lambda_-\right)$$

where $\rho(\cdot)$ denotes the spectral radius.

Proof. (Not examinable).

$$\begin{aligned} &\|\widetilde{C} - \widetilde{C}_{pos}\|_2 = \|(\widetilde{C}_{pos} - \widetilde{C}_{neg}) - \widetilde{C}_{pos}\|_2 = \| - \widetilde{C}_{neg}\|_2 = \rho(\widetilde{C}_{neg}) = \rho(\Lambda_-) \\ &\text{By Cauchy interlacing theorem} \quad \|C_N - \widehat{C}\|_2 \leq \|\widetilde{C} - \widetilde{C}_{pos}\|_2 = \rho(\Lambda_-) \end{aligned} \tag{2.66}$$

□

3. Sampling Methods for Forward UQ in ODEs

3.1. (Revision) Monte Carlo Estimate of Mean

Let $X : \Omega \to \mathbf{R}$ be a real-valued random variable with

$$\mathbf{E}[X] = \mu, \quad \mathbf{Var}[X] = \sigma^2 < \infty. \tag{3.1}$$

Let $X_1, \ldots, X_M$ be independent samples of X, i.e.,

$$X_i = X(\omega_i), \quad i = 1, 2, \ldots, M. \tag{3.2}$$

The standard MC estimate for μ is

$$\mu_M := \frac{1}{M} \sum_{i=1}^{M} X_i. \tag{3.3}$$

Recall.

- μ_M is a *random variable*, and so is the error $|\mu_M - \mu|$.
- μ_M is an *unbiased* estimate for μ since

$$\mathbf{E}[\mu_M] = \frac{1}{M}(M\mu) = \mu. \tag{3.4}$$

3.1.1. Mean-Square Error

$$\mu \approx \mu_M := \frac{1}{M}\sum_{i=1}^{M} X_i. \tag{3.5}$$

Definition 3.1 (Mean-Square Error)**.** The *mean-square error* of the estimator μ_M is

$$\mathrm{MSE}(\mu_M) = \mathbf{E}[(\mu_M - \mu)^2] = \|\mu_M - \mu\|^2_{L^2(\Omega)} \tag{3.6}$$

Proposition 3.2. *Since μ_M is an unbiased estimator,*

$$\mathrm{MSE}(\mu_M) = \mathbf{E}[(\mu_M - \mathbf{E}[\mu_M])^2] = \mathbf{Var}[\mu_M] = \frac{1}{M^2}(M\sigma^2) = \frac{\sigma^2}{M}. \tag{3.7}$$

The root mean-square error (RMSE) *is*

$$\mathbf{RMSE}(\mu_M) = \sqrt{\mathrm{MSE}(\mu_M)} = \|\mu_M - \mu\|_{L^2(\Omega)} = \frac{\sigma}{\sqrt{M}} \tag{3.8}$$

Theorem 3.3 (Chebyshev Inequality)**.** *For any $\epsilon > 0$, we have*

$$\mathbf{P}(|\mu_M - \mu| \geq M^{-1/2+\epsilon}) \leq \sigma^2 M^{-2\epsilon}. \tag{3.9}$$

As $M \to \infty$, the probability of the error being larger than $O(M^{-1/2+\epsilon})$ converges to zero.

Informally, we often say that the MC error is $O(M^{-1/2})$.

3.1.2. Estimating σ^2

We can estimate σ^2 by

$$\sigma_M^2 = \frac{1}{M-1}\sum_{i=1}^{M}(X_i - \mu_M)^2. \tag{3.10}$$

This is also an *unbiased* estimator.

3.2. IVPs & Explicit Euler Method

We consider initial value problems (IVPs) of the form:

$$\frac{\mathrm{d}\boldsymbol{u}(t)}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{u}(t)), \quad 0 < t < T, \tag{3.11}$$

with vector-valued solution

$$\boldsymbol{u}(t) = [u_1(t), \ldots, u_d(t)]^\top$$

with inital condition $\boldsymbol{u}(0) = \boldsymbol{u}_0$.

***Example* 3.4** (Population Model)**.** We introduce the following *test problem* ($d = 2$ case)

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix} = \begin{pmatrix} u_1(t)\,(1 - u_2(t)) \\ u_2(t)\,(u_1(t) - 1) \end{pmatrix}, \quad \begin{pmatrix} u_1(0) \\ u_2(0) \end{pmatrix} = \begin{pmatrix} u_{1,0} \\ u_{2,0} \end{pmatrix} \tag{3.12}$$

A typical solution to this problem shown in figure 6.

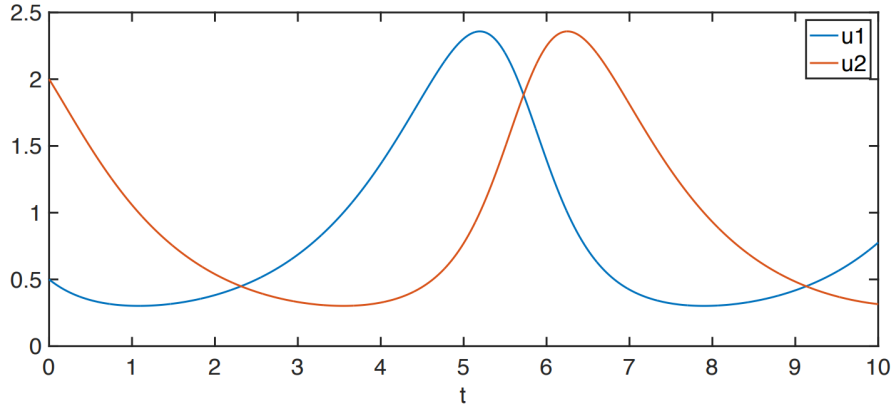

FIGURE 6: Explicit Euler approximation to the solution $\boldsymbol{u}(t) = [u_1(t), u_2(t)]^\top$ to Example 3.4 computed with $T = 10$, $N = 10^5$ and initial condition $\boldsymbol{u}_0 = [0.5, 2]^\top$.

3.2.1. Explicit Euler Method

For

$$\frac{\mathrm{d}\boldsymbol{u}(t)}{\mathrm{d}t} = \boldsymbol{f}(\boldsymbol{u}(t)), \quad 0 < t < T, \quad \boldsymbol{u}(0) = \boldsymbol{u}_0. \tag{3.13}$$

Time-stepping methods estimate the solution at a set of grid points t_n.

$$\boldsymbol{u}_n \approx \boldsymbol{u}(t_n)$$

Theorem 3.5 (Explict Euler Method)**.**

- *Divide* $[0, T]$ *into* N *intervals, and define the step size*

$$\Delta t := T/N.$$

- *Choose the* grid point $t_n = n\Delta t$, $n = 0, 1, \ldots, N$.
- *Starting with* $\boldsymbol{u}_0$, *compute*

$$\boldsymbol{u}_{n+1} = \boldsymbol{u}_n + \Delta t\ \boldsymbol{f}(\boldsymbol{u}_n), \quad n = 1, 2, \ldots, N. \tag{3.14}$$

What can we say about the *error* $\|\boldsymbol{u}(t_n) - \boldsymbol{u}_n\|_2$ at time t_n?

Theorem 3.6. *Suppose* $\boldsymbol{f} : \mathbf{R}^d \to \mathbf{R}^d$ *satisfies the Lipschitz condition*

$$\|\boldsymbol{f}(\boldsymbol{v}) - \boldsymbol{f}(\boldsymbol{w})\|_2 \le L\|\boldsymbol{x} - \boldsymbol{w}\|_2, \quad \forall \boldsymbol{v}, \boldsymbol{w} \in \mathbf{R}^d. \tag{3.15}$$

for some $L > 0$ *and that the true solution* $\boldsymbol{u}(t)$ *is twice* continuosly differentiable. *Then, for any* $\boldsymbol{u}_0 \in \mathbf{R}^d$, *there exist a constant* $K > 0$ *independent of* Δt *such that*

$$\max_{0 \le t_n \le T} \|\boldsymbol{u}(t_n) - \boldsymbol{u}_n\|_2 \le K\Delta t. \tag{3.16}$$

Hence the explicit Euler error is $O(\Delta t)$.

3.3. IVP & Uncertain Initial Condition

***Example* 3.7** (Population Model)**.**

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{pmatrix} u_1(t) \\ u_2(t) \end{pmatrix} = \begin{pmatrix} u_1(t)\,(1 - u_2(t)) \\ u_2(t)\,(u_1(t) - 1) \end{pmatrix}, \quad \begin{pmatrix} u_1(0) \\ u_2(0) \end{pmatrix} = \begin{pmatrix} u_{1,0} \\ u_{2,0} \end{pmatrix} \tag{3.17}$$

If we model $u_{1,0}$ and $u_{2,0}$ as *random variables*, then the ODE solution $\boldsymbol{u} : [0,T] \times \Omega \to \mathbf{R}^2$ is a vector-valued *stochastic process*

$$\boldsymbol{u}(t,\omega) = \begin{pmatrix} u_1(t,\omega) \\ u_2(t,\omega) \end{pmatrix}. \tag{3.18}$$

For this example, we modelled as

$$u_{1,0} \sim U(\overline{u}_1 - \tau, \overline{u}_1 + \tau), \qquad u_{2,0} \sim U(\overline{u}_2 - \tau, \overline{u}_2 + \tau). \tag{3.19}$$

and we choose: $\overline{u}_1 = 0.5$, $\overline{u}_2 = 2$, and $\tau = 0.1$. For each sample $\boldsymbol{u}_0^i = u_0(\omega_i)$ of the initial condition, we can apply the explicit Euler method (with N intervals) to approximate

$$\boldsymbol{u}_n^i \approx \boldsymbol{u}(t_n, \omega_i), \quad n = 1, 2, \ldots, N, \tag{3.20}$$

at the grid points t_n on $[0,T]$. If we choose $T = 6$ and $N = 10^6$, we have the figure 7.

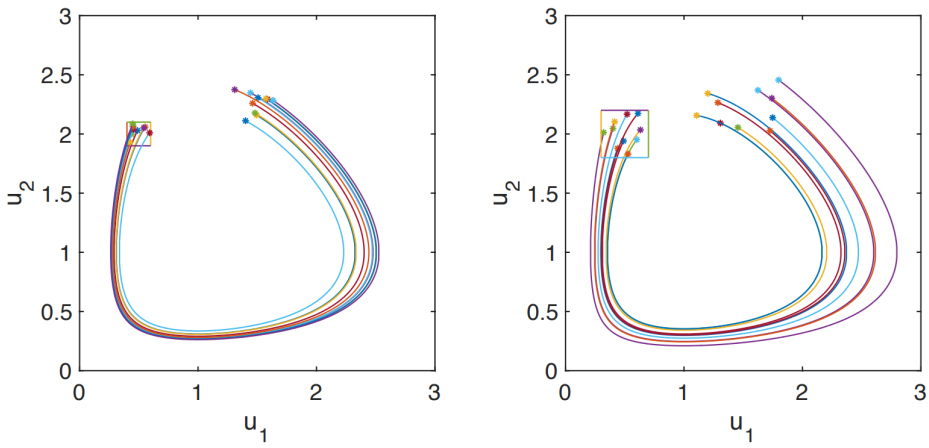

FIGURE 7: Ten samples of the explicit Euler approximation to the solution to Example 3.7 computed with $T = 6$, and $N = 10^6$ and $\tau = 0.1$ (left) and $\tau = 0.2$ (right).

3.3.1. Estimating Mean of Quantity of Intest

Suppose our quantity of interest is

$$X := u_2(T, \omega). \tag{3.21}$$

Key Question. How can we estimate $\mathbf{E}[X]$ efficiently and accurately?

But we don't have access to samples of the exact solution $X_i := u_2(T, \omega_i)$. For a fixed N, we can define the random variable

$$X_N = u_{2,N}, \quad \text{where } \boldsymbol{u}_N(\omega) = \begin{pmatrix} u_{1,N}(\omega) \\ u_{2,N}(\omega) \end{pmatrix} \tag{3.22}$$

is the *explicit Euler* approximation at $t_N = T$.

Let X_N^i denote a sample of X_N and define the Monte Carlo estimate

$$\mu_{M,N} := \frac{1}{M} \sum_{i=1}^{M} X_N^i \approx \mathbf{E}[X_N] \approx \mathbf{E}[X]. \tag{3.23}$$

If we use $\mu_{M,N}$ to estimate $\mathbf{E}[X]$, then the error depends on:

- the number of *samples* M,
- the number of *intervals* N (equivalently, the *step size* Δt).

3.4. Standard Monte Carlo: Cost & Accuracy

Let X be a $\mathbf{R}$-valued random variable with mean

$$\mathbf{E}[X] = \mu. \tag{3.24}$$

Let Equation (3.23) be the standard MC estimate.

Question. How do we choose M, N so that we can guarantee

$$\mathbf{RMSE}(\mu_{M,N}) \leq \epsilon. \tag{3.25}$$

and what will be the corresponding cost?

Theorem 3.8. *Let* $\mu_{M,N} := \frac{1}{M}\sum_{i=1}^{M} X_N^i \approx \mu$, *then*

$$\mathrm{MSE}(\mu_{M,N}) = \mathbf{E}[(\mu_{M,N} - \mu)^2] = \underbrace{M^{-1}\mathbf{Var}[X_N]}_{\textit{MC Samping Error}} + \underbrace{(\mathbf{E}[X_N - X])^2}_{\textit{Bias term } \mathbf{E}[X_N] \neq \mathbf{E}[X]} \tag{3.26}$$

Proof.

$$\begin{aligned}
\mathrm{MSE}(\mu_{M,N}) &= \mathbf{E}[\mu_{M,N} - 2\mu\mu_{M,N} + \mu^2] \\
&= \mathbf{E}[\mu_{M,N}^2] - 2\mu\,\mathbf{E}[\mu_{M,N}] + \mu^2 \\
&= \mathbf{E}[\mu_{M,N}^2] - \mathbf{E}[\mu_{M,N}]^2 + \mathbf{E}[\mu_{M,N}]^2 - 2\mu\,\mathbf{E}[\mu_{M,N}] + \mu^2 \\
&= \mathbf{Var}[\mu_{M,N}] + (\mathbf{E}[\mu_{M,N}] - \mu)^2 \\
&= \mathbf{Var}[\mu_{M,N}] + (\mathbf{E}[\mu_{M,N} - X])^2 \quad (\mu = \mathbf{E}[X]).
\end{aligned}$$

$$\begin{aligned}
\mathbf{Var}[\mu_{M,N}] &= \mathbf{Var}[\frac{1}{M}\sum_{i=1}^{M} X_N^i] \\
&= \frac{1}{M^2}\sum_{i=1}^{M}\mathbf{Var}[X_N^i] = \frac{1}{M}\mathbf{Var}[X_N].
\end{aligned} \tag{3.27}$$

$$\begin{aligned}
\mathbf{E}[\mu_{M,N}] &= \frac{1}{M}\sum_{i=1}^{M}\mathbf{E}[X_N^i] = \mathbf{E}[X_N] \\
\mathrm{MSE}(\mu_{M,N}) &= M^{-1}\mathbf{Var}[X_N] + (\mathbf{E}[X_N - X])^2 \quad \text{as required.}
\end{aligned}$$

□

3.4.1. Bias : Accuracy of X_N

To investigate the *bias*, we need information about the quality of the approximation, $X_N \approx X$, and how the error depends on N.

Assumption 3.9. *We assume that* $\exists C$ *and* $\alpha > 0$ *such that*

$$|\,\mathbf{E}[X_N - X]| \leq CN^{-\alpha}, \tag{3.28}$$

where C is independent of N.

- *X_N converges* in mean *to X as $N \to \infty$,*
- *α is the rate of convergence.*

3.4.2. Cost to Controlling the MSE

To obtain $\mathbf{RMSE}(\mu_{M,N}) \leq \epsilon$, where ϵ is a fixed tolerance, we need $\text{MSE}(\mu_{M,N}) \leq \epsilon^2$. From Theorem 3.8, this is satisfied if

$$M^{-1}\mathbf{Var}[X_N] \leq \frac{\epsilon^2}{2} \qquad \& \qquad (\mathbf{E}[X_N - X])^2 \leq \frac{\epsilon^2}{2} \tag{3.29}$$

Hence we could choose

- $M \geq 2\mathbf{Var}[X_N]\epsilon^{-2} \to M = O(\epsilon^{-2})$.
- $N \geq (2C^2)^{1/2\alpha}\epsilon^{-1/\alpha} \to N = O(\epsilon^{-1/\alpha})$. (From Assumption 3.9)

Theorem 3.10 (Cost of MC estimator)**.** *If the cost of generating one sample of X_N is $O(N^\beta)$, then the cost to compute $\mu_{M,N}$ is*

$$C_{MC} = M \times O(N^\beta). \tag{3.30}$$

The cost to deliver an estimate such that $\mathbf{RMSE}(\mu_{M,N}) \leq \epsilon$ is

$$C_{MC} = O(\epsilon^{-2}) \times O(\epsilon^{-\beta/\alpha}) = O(\epsilon^{-2-\beta/\alpha}). \tag{3.31}$$

3.5. Multilevel Monte Carlo : Key idea

Recap: Let X be real-valued r.v. with mean $\mathbf{E}[X] = \mu$. Suppose $X_N \approx X$ is another r.v. associated with a *discretization parameter* N and the standard Monte Carlo estimate is:

$$\mu_{M,N} := \frac{1}{M}\sum_{i=1}^{M} X_N^i \approx \mu$$

with the *mean-square error*

$$\text{MSE}(\mu_{M,N}) = M^{-1}\underbrace{\mathbf{Var}[X_N]}_{\text{Sampling error}} + (\underbrace{\mathbf{E}[X_N - X]}_{\text{bias due to approximation}})^2$$

But we have some key disadvantages:

1. Error depends on $\mathbf{Var}[X_N]$.
2. Cost to achieve $\mathbf{RMSE} \leq \epsilon$ often too high.

3.5.1. Sequence of Approximation

Suppose we have access to samples of a sequence of r.v.s $X_{N_0}, X_{N_1}, \ldots, X_{N_L}$ that each approximate X, where N_0 is fixed and

$$N_\ell = sN_{\ell-1}, \quad \ell = 1, 2, \ldots, L,$$

for some $s > 1$. (We ensure that $N_0 < N_1 < \cdots < N_L$)

Here ℓ is the *level* number of the approximation X_{N_ℓ}.

Assumption 3.11.

- *X_{N_ℓ} is a more* accurate *approxoimation of X as $\ell \to \infty$.*
- *Samples of X_{N_ℓ} Become more* expensive *to generate as $\ell \to \infty$.*

3.5.2. MLMC : Basic Ideas

In the MLMC method, we use a *decreasing* number of samples of X_{N_ℓ} to estimate $\mu = \mathbf{E}[X]$ as the level number ℓ *increases*.

Recall. Samples of X_{N_0} are cheaper to compute and samples of X_{N_ℓ} are the most *expensive* to compute.

3.6. MLMC Estimator : Definition & Properties

The standard MC estimator of $\mu = \mathbf{E}[X]$ associated with level L is

$$\mu_{M,N_L} := \frac{1}{M}\sum_{i=1}^{M} X_{N_L}^i \approx \mu.$$

3.6.1. Telescoping Sum

First observe that $\mathbf{E}[X_{N_L}]$ can be decomposed as follows:

$$\mathbf{E}[X_{N_L}] = \mathbf{E}[X_{N_0}] + \sum_{\ell=1}^{L}\left\{\mathbf{E}[X_{N_\ell}] - \mathbf{E}[X_{N_{\ell-1}}]\right\},$$

using linearity of $\mathbf{E}[\cdot]$, we have

$$\mathbf{E}[X_{N_L}] = \mathbf{E}[X_{N_0}] + \sum_{\ell=1}^{L}\underbrace{\mathbf{E}[X_{N_\ell} - X_{N_{\ell-1}}]}_{\text{Correction terms}}.$$

By defining

$$Y_0 = X_{N_0} \quad \text{and} \quad Y_\ell := X_{N_\ell} - X_{N_{\ell-1}} \quad \text{for} \quad \ell = 1, 2, \ldots, L.$$

We could write

$$\mathbf{E}[X_{N_L}] = \mathbf{E}[Y_0] + \sum_{\ell=1}^{L}\mathbf{E}[Y_\ell] \to \text{The Telescoping Sum}$$

3.6.2. MLMC Estimator

Idea: Estimate the terms associated with each level *separately* using the standard MC method with M_ℓ samples.

1. On *level* 0, define
$$\widehat{\mu_0} := \frac{1}{M_0}\sum_{i=1}^{M_0} Y_0^i,$$
using M_0 *independent* samples of $Y_0 = X_{N_0}$.
2. On *level* $1, 2, \ldots, L$, define
$$\widehat{\mu_\ell} := \frac{1}{M_\ell}\sum_{i=1}^{M_\ell} Y_\ell^i,$$
using M_ℓ *independent* samples of $Y_\ell := X_{N_\ell} - X_{N_{\ell-1}}$.

NOTE. To stress that samples should be *independent* for each level ℓ, we can also use the alternative notation:

$$\widehat{\mu}_\ell := \frac{1}{M_\ell}\sum_{i_\ell=1}^{M_\ell} Y_{N_\ell}^{i_\ell}.$$

Using the definition of Y_{N_ℓ} gives,

$$Y^i_{N_\ell} = X^i_{N_\ell} - X^i_{N_{\ell-1}} = X_{N_\ell}(\omega_i) - X_{N_{\ell-1}}(\omega_i)$$

To compute $\widehat{\mu_\ell}$, $X^i_{N_\ell}$ and $X^i_{N_{\ell-1}}$ correspond to the same input sample.

Combing the estimates for each term, the final MLMC estimator is

$$\widehat{\mu}_{\text{MLMC}} := \sum_{\ell=0}^{L} \widehat{\mu_\ell}$$

Question.

1. How to choose L? (No. of levels)
2. How to choose $M_0, M_1, \ldots, M_L$? (No. of samples at each level)

Theorem 3.12 (Mean and Variance of MLMC estimator)**.**

$$\begin{aligned} \mathbf{E}[\widehat{\mu}_{\text{MLMC}}] &= \mathbf{E}[\widehat{\mu_L}] = \mathbf{E}[X_{N_L}], \\ \mathbf{Var}[\widehat{\mu}_{\text{MLMC}}] &= \sum_{\ell=0}^{L} M_\ell^{-1}\mathbf{Var}[Y_\ell] \end{aligned} \tag{3.32}$$

Proof.

$$\mathbf{E}[\widehat{\mu}_{\text{MLMC}}] = \mathbf{E}[\sum_{\ell=0}^{L} \widehat{\mu}_\ell] = \sum_{\ell=0}^{L} \mathbf{E}[\widehat{\mu}_\ell].$$

We evaluate these separately

$$\begin{aligned} \mathbf{E}[\widehat{\mu}_0] &= \mathbf{E}[\frac{1}{M_0}\sum_{i=1}^{M_0} X^i_{N_0}] = \frac{1}{M_0}\sum_{i=1}^{M_0}\mathbf{E}[X^i_{N_0}] = \mathbf{E}[X_{N_0}]. \\ \mathbf{E}[\widehat{\mu}_\ell] &= \cdots = \mathbf{E}[X_{N_\ell}] - \mathbf{E}[X_{N_{\ell-1}}]. \end{aligned}$$

Substituted into the first equation:

$$\mathbf{E}[\widehat{\mu}_{\text{MLMC}}] = \mathbf{E}[X_{N_0}] + \mathbf{E}[X_{N_1}] - \mathbf{E}[X_{N_0}] + \cdots + \mathbf{E}[X_{N_L}] - \mathbf{E}[X_{N_{L-1}}] = \mathbf{E}[X_{N_L}]$$

$$\mathbf{Var}[\widehat{\mu}_{\text{MLMC}}] = \mathbf{Var}[\sum_{\ell=0}^{L} \widehat{\mu}_\ell] = \sum_{\ell=0}^{L} \mathbf{Var}[\widehat{\mu}_\ell].$$

For $\ell = 0, 1, 2 \ldots, L$, we always have

$$\mathbf{Var}[\widehat{\mu}_\ell] = \mathbf{Var}[\frac{1}{M_\ell}\sum_{i=1}^{M_\ell} Y_\ell] = \frac{1}{M_\ell^2}\sum_{i=1}^{M_\ell}\mathbf{Var}[Y_\ell] = M_\ell^{-1}\mathbf{Var}[Y_\ell]$$

Substituted into the first equation

$$\mathbf{Var}[\widehat{\mu}_{\text{MLMC}}] = \sum_{\ell=0}^{L}\mathbf{Var}[\widehat{\mu}_\ell] = \sum_{\ell=0}^{L} M_\ell^{-1}\mathbf{Var}[Y_\ell].$$

□

3.7. MLMC : Mean Square Error

Goal: Compute an estimator $\widehat{\mu}_{\text{MLMC}}$ such that

$$\mathbf{RMSE}(\widehat{\mu}_{\text{MLMC}}) \leq \epsilon \iff \text{MSE}(\widehat{\mu}_{\text{MLMC}}) \leq \epsilon^2$$

Theorem 3.13 (Mean Square Error of MLMC estimator)**.** *The mean-square error of the MLMC estimator $\widehat{\mu}_{\text{MLMC}}$ satisfies*

$$\text{MSE}(\widehat{\mu}_{\text{MLMC}}) = \sum_{\ell=0}^{L} M_\ell^{-1}\mathbf{Var}[Y_\ell] + (\mathbf{E}[X_{N_L} - X])^2$$

Proof.

$$\begin{aligned}
\text{MSE}(\widehat{\mu}_{\text{MLMC}}) &= \mathbf{E}\left[(\widehat{\mu}_{\text{MLMC}} - \mu)^2\right] \\
&= \mathbf{E}\left[((\widehat{\mu}_{\text{MLMC}} - \mathbf{E}\left[\widehat{\mu}_{\text{MLMC}}\right]) + (\mathbf{E}\left[\widehat{\mu}_{\text{MLMC}}\right] - \mu))^2\right] \\
&= \mathbf{E}\left[\left(\underbrace{\widehat{\mu}_{\text{MLMC}} - \mathbf{E}\left[\widehat{\mu}_{\text{MLMC}}\right]}_{\text{R.V. with mean 0}}\right)^2\right] + \mathbf{E}\left[\left(\underbrace{\mathbf{E}\left[\widehat{\mu}_{\text{MLMC}}\right] - \mu}_{\text{real number}}\right)^2\right] \\
&\qquad - 2 \cdot \underbrace{\mathbf{E}\left[(\widehat{\mu}_{\text{MLMC}} - \mathbf{E}\left[\widehat{\mu}_{\text{MLMC}}\right])\left(\mathbf{E}\left[\widehat{\mu}_{\text{MLMC}}\right] - \mu\right)\right]}_{=0} \\
&= \mathbf{E}\left[(\widehat{\mu}_{\text{MLMC}} - \mathbf{E}\left[\widehat{\mu}_{\text{MLMC}}\right])^2\right] + (\mathbf{E}\left[\widehat{\mu}_{\text{MLMC}}\right] - \mu)^2 \\
&= \mathbf{Var}[\widehat{\mu}_{\text{MLMC}}] + (\mathbf{E}\left[X_{N_L}\right] - \mathbf{E}[X])^2 \\
&= \mathbf{Var}[\widehat{\mu}_{\text{MLMC}}] + (\mathbf{E}\left[X_{N_L} - X\right])^2 \\
&= \sum_{\ell=0}^{L} M_\ell^{-1}\mathbf{Var}[Y_\ell] + (\mathbf{E}[X_{N_L} - X])^2
\end{aligned}$$

□

3.7.1. Controlling the MSE : Bias

Goal to achieve $\text{MSE}(\widehat{\mu}_{\text{MLMC}}) \leq \epsilon^2$, and this is satisfied if

$$(\mathbf{E}[X_{N_L} - X])^2 \leq \frac{\epsilon^2}{2} \quad \& \quad \left(\sum_{\ell=0}^{L} M_\ell^{-1}\mathbf{Var}[Y_\ell]\right) \leq \frac{\epsilon^2}{2}$$

Assumption 3.14. *We assume that there exist C and $\alpha > 0$ such that*

$$|\mathbf{E}\left[X_{N_L} - X\right]| \leq C N_L^{-\alpha},$$

where C is independent of N_L.

Using this assumption, we obtain

$$(\mathbf{E}\left[X_{N_L} - X\right])^2 \leq \frac{\epsilon^2}{2},$$

if we choose

$$N_L > (2C^2)^{1/2\alpha}\epsilon^{-1/\alpha}$$

Assuming that N_0 and s are given, and

$$N_\ell = sN_{\ell-1} \quad \text{for } \ell = 1, 2, \ldots, L.$$

How many levels L are needed to control the bias?

3.8. Variance of Correction Terms

Recall: The mean square error of MLMC estimator $\widehat{\mu}_{\text{MLMC}}$ satisfies

$$\text{MSE}(\widehat{\mu}_{\text{MLMC}}) = \sum_{\ell=0}^{L} M_\ell^{-1}\mathbf{Var}[Y_\ell] + \underbrace{(\mathbf{E}[X_{N_L} - X])^2}_{\text{bias term}}$$

The bias term can be satisfied if we choose L properly by section 1.3.1. How we could deal with

$$\left(\sum_{\ell=0}^{L} M_\ell^{-1}\mathbf{Var}[Y_\ell]\right) \leq \frac{\epsilon^2}{2}.$$

If we have a *sequence* of approximations $X_{N_0}, X_{N_1}, \ldots, X_{N_L}$ to X and $Y_0 = X_{N_0}$, and $Y_\ell := X_{N_\ell} - X_{N_{\ell-1}}$ for $\ell = 1, 2, \ldots, L$.

Assumption 3.15. *The sequence $X_{N_0}, X_{N_1}, \ldots, X_{N_L}$ converges to X in the* mean-square *sense, i.e.*

$$\mathbf{E}\left[(X_{N_\ell} - X)^2\right] = \|X_{N_\ell} - X\|^2_{L^2(\Omega)} \to 0 \quad \textit{as} \quad \ell \to \infty.$$

Lemma 3.16. *If the assumption 3.15 is satisfied, then*

$$\mathbf{Var}[Y_\ell] \to 0 \quad \textit{as} \quad \ell \to \infty$$

Proof. For $\ell = 1, 2, \ldots$, (not include 0).

$$\begin{aligned}
\mathbf{Var}[Y_\ell] &= \mathbf{E}[Y_\ell^2] - (\mathbf{E}[Y_\ell])^2 \\
&\leq \mathbf{E}[Y_\ell^2] \\
&= \|Y_\ell\|^2_{L^2(\Omega)} \\
&= \left\|X_{N_\ell} - X_{N_{\ell-1}}\right\|^2_{L^2(\Omega)} \\
&= \left\|X_{N_\ell} - X + X - X_{N_{\ell-1}}\right\|^2_{L^2(\Omega)} \\
&\leq \left(\underbrace{\|X_{N_\ell} - X\|_{L^2(\Omega)}}_{\to 0 \text{ as } \ell\to\infty} + \underbrace{\left\|X - X_{N_{\ell-1}}\right\|_{L^2(\Omega)}}_{\to 0 \text{ as } (\ell-1)\to\infty}\right)^2 \\
&\Rightarrow \mathbf{Var}[Y_\ell] \to 0 \ \text{ as } \ \ell \to \infty
\end{aligned}$$

□

3.8.1. Mean Square Error

We need to choose $M_0, M_1, \ldots, M_L$ such that

$$M_0^{-1}\mathbf{Var}[X_{N_0}] + \sum_{\ell=1}^{L} M_\ell^{-1}\mathbf{Var}[Y_\ell] \leq \frac{\epsilon^2}{2}.$$

Simple way: There are $L+1$ terms in total. The condition is satisfied if

$$M_0^{-1}\mathbf{Var}[X_{N_0}] \leq \frac{\epsilon^2}{2(L+1)} \qquad \& \qquad M_\ell^{-1}\mathbf{Var}[Y_\ell] \leq \frac{\epsilon^2}{2(L+1)}.$$

That is, if we choose the *number of samples* such that

$$M_0 \geq 2(L+1)\mathbf{Var}[X_{N_0}]\epsilon^{-2} \qquad \& \qquad M_\ell \geq 2(L+1)\mathbf{Var}[Y_\ell]\epsilon^{-2}$$

Key Conclusion: If $\mathbf{Var}[Y_\ell] \to 0$ as $\ell \to \infty$, the number of samples M_ℓ we need to choose to control the MSE also decrease as $\ell \to \infty$.

3.9. MLMC : Accuracy and Cost

Recall. MLMC estimator for $\mu = \mathbf{E}[X]$ is

$$\widehat{\mu}_{\text{MLMC}} := \sum_{\ell=0}^{L} \widehat{\mu}_\ell \quad \text{where } \widehat{\mu}_\ell = \frac{1}{M_\ell} \sum_{i=1}^{M_\ell} Y_\ell^i$$

where we define the following:

$$Y_0 := X_{N_0}, \quad Y_\ell := X_{N_\ell} - X_{N_{\ell-1}}, \quad \ell = 1, 2, \ldots, L.$$

Question. What is the *cost* of computing $\widehat{\mu}_{\text{MLMC}}$ in such a way that $\mathbf{RMSE}(\widehat{\mu}_{\text{MLMC}}) \leq \epsilon$

3.9.1. MLMC Cost

Let C_{N_ℓ} denote the cost of generate *one* sample of Y_ℓ, $\ell = 0, 1, \ldots, L$. Hence the total cost of computing the samples needed for $\widehat{\mu}_{\text{MLMC}}$ is

$$C_{MLMC} = \sum_{\ell=0}^{L} M_\ell \times C_{N_\ell}. \tag{3.33}$$

***Example* 3.17** (Population Model)**.** The cost of computing one sample of X_{N_ℓ} using the explicit Euler method with N_ℓ intervals can be *characteristised by the number of intervals.*

$$C_{N_0} = N_0, \quad C_{N_\ell} = N_\ell + N_{\ell-1} = (1 + s^{-1})N_\ell = (1 + s^{-1})s^\ell N_0. \tag{3.34}$$

Hence,

$$C_{MLMC} = M_0 N_0 + \sum_{\ell=1}^{L} M_\ell (1 + s^{-1}) s^\ell N_0. \tag{3.35}$$

The cost dependents on s, N_0 and L and M_ℓ.

3.9.2. Controlling the MSE

Recall (Theorerm 3.13)**.** The mean-square error of $\widehat{\mu}_{\text{MLMC}}$ satisfies

$$\text{MSE}(\widehat{\mu}_{\text{MLMC}}) = \sum_{\ell=0}^{L} M_\ell^{-1} \mathbf{Var}[Y_\ell] + (\mathbf{E}[X_{N_\ell} - X])^2.$$

To satisfy $\text{MSE}(\widehat{\mu}_{\text{MLMC}}) \leq \epsilon^2$, we could choose L (no. of levels) and M_ℓ (no. of samples at each levels) such that

$$(\mathbf{E}[X_{N_\ell} - X])^2 \leq \frac{\epsilon^2}{2} \quad \& \quad \sum_{\ell=0}^{L} M_\ell^{-1} \mathbf{Var}[Y_\ell] \leq \frac{\epsilon^2}{2}. \tag{3.36}$$

3.9.3. Number of Samples.

If we choose the number of samples so that

$$M_0 \geq 2(L+1)\mathbf{Var}[X_{N_0}]\epsilon^{-2} \quad \& \quad M_\ell \geq 2(L+1)\mathbf{Var}[Y_\ell]\epsilon^{-2}. \tag{3.37}$$

to get integers:

$$M_0 = \lceil 2(L+1)\mathbf{Var}[X_{N_0}]\epsilon^{-2} \rceil \quad \& \quad \lceil M_\ell = 2(L+1)\mathbf{Var}[Y_\ell]\epsilon^{-2} \rceil.$$

***Example* 3.18** (Population Model)**.** We have proved that $\mathbf{Var}[Y_\ell] \leq C_s T^2 N_\ell^{-2}$, hence we could choose $M_\ell = \lceil 2(L+1) C_s T^2 N_\ell^{-2} \epsilon^{-2} \rceil$

3.9.4. Number of Levels.

Assumption 3.19. *There exist C and $\alpha > 0$ such that*

$$|\,\mathbf{E}[X_{N_\ell} - X]| \leq C N_L^{-\alpha},$$

where C is independent of L.

Then we obtain $(\mathbf{E}[X_{N_\ell} - X])^2 \leq \epsilon^2/2$ if we choose $N_L \geq (2C^2)^{1/2\alpha}\epsilon^{-1/\alpha}$.

***Example* 3.20** (Population Model ($\alpha = 1$))**.** Since $N_L = s^L N_0$, we show that $L = O(|\log(\epsilon)|)$. Then the *total cost of the Population model* will be

$$C_{MLMC} := \sum_{\ell=0}^{L} M_\ell \times C_{N_\ell} = M_0 N_0 + \sum_{\ell=1}^{L} M_\ell (1 + s^{-1}) N_\ell.$$

We obtain $\mathrm{MSE}(\widehat{\mu}_{\mathrm{MLMC}}) \leq \epsilon^2$ if we choose:

- $L = O(|\log(\epsilon)|)$.
- $M_0 = \lceil 2(L+1)\mathbf{Var}[X_{N_0}]\epsilon^{-2}\rceil$.
- $M_\ell = \lceil 2(L+1)C_s T^2 N_\ell^{-2}\epsilon^{-2}\rceil$.

Since M_ℓ decays to zero more quickly than C_{N_ℓ} increase as $\ell \to \infty$, the dominant term is

$$M_0 N_0 = \lceil 2(L+1)\mathbf{Var}[X_{N_0}]\epsilon^{-2}\rceil N_0 = O(\epsilon^{-2}|\log(\epsilon)|) \tag{3.38}$$

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